Communication Complexity

Lecture Notes

Adapted from lecture notes taken in the course
Communication Complexity, taught by Professor Hsueh-I Lu
National Taiwan University, Spring 2026

Yu-Sheng Shih

October 2026

Abstract. These notes give an introduction to two-party communication complexity. Part I develops deterministic communication: protocol trees and combinatorial rectangles, fooling sets, rectangle covers and nondeterminism, and the rank method together with the log-rank conjecture. Part II studies randomized protocols, Newman's theorem, Yao's minimax principle and the discrepancy method. Part III turns to the direct-sum question and develops the information-theoretic tools that lead to the $\Omega(n)$ lower bound on the randomized communication complexity of set disjointness.

These notes were edited and integrated with the assistance of GPT (OpenAI) and Claude Opus 5.5 (Anthropic).

Contents

Preface **iv**

Notation **vi**

I Deterministic Communication **1**

1 The Two-Party Communication Model **2**
1.1 Communication games . . . 2
1.2 Example: the median . . . 3
1.3 Example: clique versus independent set . . . 5
1.4 Notes and references . . . 5

2 Protocol Trees and Combinatorial Rectangles **7**
2.1 Protocol trees . . . 7
2.2 Rectangles . . . 8
2.3 Fooling sets . . . 8
2.4 The rectangle-size bound . . . 9
2.5 Counting leaves and balancing protocols . . . 9
2.6 From monochromatic partitions to protocols . . . 11
2.7 Notes and references . . . 12

3 Nondeterministic Communication **13**
3.1 Rectangle covers . . . 13
3.2 Nondeterministic protocols . . . 13
3.3 Certificates on both sides . . . 14
3.4 Notes and references . . . 15

4 Rank and the Log-Rank Conjecture **16**
4.1 Rank, nonnegative rank and the rank lower bound . . . 16
4.2 Examples . . . 16
4.3 Upper bounds from rank . . . 17
4.4 Covers and Lovász's bound . . . 17
4.5 The log-rank conjecture . . . 19
4.6 Notes and references . . . 20

II Randomized Communication **21**

5 Randomized Protocols **22**
5.1 Private and public coins . . . 22
5.2 Equality . . . 23
5.3 Greater-than . . . 24
5.4 Newman's theorem: public versus private coins . . . 25
5.5 Private coins save at most exponentially . . . 26
5.6 Summary . . . 26
5.7 Notes and references . . . 27

6 Distributional Complexity and Yao's Principle **28**
6.1 Distributional complexity . . . 28
6.2 The minimax theorem . . . 29
6.3 Yao's minimax principle . . . 29
6.4 Notes and references . . . 30

7 The Discrepancy Method **31**
7.1 Discrepancy and the lower bound . . . 31
7.2 Inner product . . . 32
7.3 Greater-than . . . 32
7.4 Limitations, and a summary . . . 33
7.5 Notes and references . . . 33

III Many Copies and Information **35**

8 Many Copies: Direct Sums and Amortization **36**
8.1 Definitions . . . 36
8.2 Upper bounds . . . 36
8.3 Relations: batching can help a lot . . . 37
8.4 A direct-sum lower bound for functions . . . 37
8.5 Notes and references . . . 39

9 Entropy, Divergence and Mutual Information **40**
9.1 Self-information and prefix-free codes . . . 40
9.2 Entropy . . . 40
9.3 Divergence and Pinsker's inequality . . . 41
9.4 Random variables and conditional entropy . . . 42
9.5 Mutual information . . . 43
9.6 Notes and references . . . 45

10 Information Complexity and the Hardness of Disjointness **46**
10.1 Hellinger distance . . . 46
10.2 Warm-up: the one-way index problem . . . 47
10.3 Information cost . . . 47
10.4 The hard distribution . . . 48
10.5 The direct-sum step . . . 49
10.6 The one-bit lower bound . . . 50
10.7 Disjointness . . . 50
10.8 The big picture . . . 51
10.9 Notes and references . . . 51

A Probabilistic and Analytic Tools **52**
A.1 Elementary inequalities . . . 52
A.2 Concentration . . . 52

Bibliography **53**

Preface

Communication complexity studies a deceptively simple question: two parties, Alice and Bob, each hold half of the input of a function $f(x, y)$; how many bits must they exchange in order to compute f? The model ignores computation entirely and charges only for communication. This makes it simple enough that we can actually prove strong lower bounds, and yet general enough that those lower bounds transfer to circuits, streaming algorithms, data structures, extended formulations of polytopes and many other models [KN97, RY20].

These notes are an introduction to the subject for readers who meet it for the first time. The only prerequisites are discrete probability, basic linear algebra (rank, eigenvalues, the Cauchy–Schwarz inequality) and asymptotic notation; the probabilistic inequalities that are used repeatedly are collected in Appendix A.

How these notes came about. The notes grew out of the author's notes for a graduate course on communication complexity taught by Professor Hsueh-I Lu at National Taiwan University in the spring semester of 2026. Those notes were later edited into a first typeset version and then reorganised and rewritten into the present self-contained text: the order of topics was rearranged so that each part tells one story, arguments that had only been sketched were completed, and every chapter received a section of bibliographic notes.

What the notes cover. The notes are organised in three parts.

- **Part I (Deterministic communication).** The model and two surprising protocols (Chapter 1); protocol trees and combinatorial rectangles, with lower bounds by fooling sets and rectangle size, and upper bounds by balancing protocol trees and from monochromatic partitions (Chapter 2); rectangle covers and nondeterministic protocols, culminating in the Aho–Ullman–Yannakakis theorem $\mathrm{D}(f) = O(\mathrm{N}^0(f)\mathrm{N}^1(f))$ (Chapter 3); and the rank method, Lovász's bound and the log-rank conjecture (Chapter 4).
- **Part II (Randomized communication).** Private and public coins, protocols for equality and greater-than, Newman's theorem (Chapter 5); distributional complexity and Yao's minimax principle (Chapter 6); the discrepancy method, with applications to inner product and greater-than (Chapter 7).
- **Part III (Many copies and information).** The direct-sum question: what does it cost to solve many independent copies of a problem (Chapter 8); the basic notions of information theory (Chapter 9); and their use in the one-way index lower bound and in the $\Omega(n)$ randomized lower bound for set disjointness (Chapter 10).

Chapters 1 and 2 are needed for everything that follows. After them, Part II can be read before or after the rest of Part I. Chapter 8 uses Chapter 3 and Chapter 5; Chapter 9 is self-contained; and Chapter 10 uses Chapters 5, 6 and 9. Each chapter ends with a section of *Notes and references* pointing to the original papers and to the relevant chapters of the textbooks of Kushilevitz and Nisan [KN97] and of Rao and Yehudayoff [RY20], which are the natural next step for the reader.

Conventions. All logarithms are to base 2 unless written ln. For a positive integer n we write $[n] = \{1, \dots, n\}$. We freely identify a string $x \in \{0,1\}^n$ with the subset $\{i : x_i = 1\} \subseteq [n]$ and, when convenient, with the integer $\sum_{i=1}^{n} x_i 2^{n-i}$ (so x_1 is the most significant bit). The notation for the complexity measures is summarised on the next page.

Acknowledgments. The author thanks Professor Hsueh-I Lu for the course from which these notes originated. The text was edited and integrated with the assistance of AI language models: a first typeset version of the original notes was prepared with OpenAI's GPT, and the present version was reorganised, rewritten and cross-checked against the literature with Anthropic's Claude Opus 5.5. As with any notes, errors may remain; the cited textbooks and papers are the authoritative sources.

Notation

We follow the notation of Kushilevitz and Nisan [KN97] for the main complexity measures, and standard information-theoretic notation otherwise [CT06]. Because different papers use different symbols, the last column lists some common alternatives.

Symbol	Meaning	Also written
$\mathrm{D}(f)$	deterministic communication complexity	$D^{\mathrm{cc}}(f)$, $CC(f)$
$C^D(f)$	partition number: min. size of a monochromatic partition	$\chi(f)$ [GPW15]
$C^P(f)$	protocol partition number: min. number of leaves of a protocol tree	—
$C^1(f)$, $C^0(f)$	min. size of a monochromatic 1-cover / 0-cover	—
$C(f)$	min. size of a monochromatic cover of $X \times Y$	—
$\mathrm{R}_\varepsilon(f)$	private-coin randomized complexity, error ε	$R^{\mathrm{priv}}_\varepsilon(f)$
$\mathrm{R}^{\mathrm{pub}}_\varepsilon(f)$	public-coin randomized complexity, error ε	—
$\mathrm{R}(f)$, $\mathrm{R}^{\mathrm{pub}}(f)$	the same with $\varepsilon = 1/3$	—
$\mathrm{D}^\mu_\varepsilon(f)$	distributional complexity	—
$\mathrm{disc}_\mu(f)$	discrepancy	—
$\mathrm{N}^1(f)$, $\mathrm{N}^0(f)$	nondeterministic complexity	—
$\mathrm{IC}_\mu(\Pi)$	(internal) information cost of a protocol	$\mathrm{IC}^{\mathrm{int}}_\mu(\Pi)$
$I(X;Y)$, $I(X;Y \mid Z)$	mutual information, conditional mutual information	$I(X:Y)$
$D_{\mathrm{KL}}(\pi \,\|\, \tau)$	Kullback–Leibler divergence	$D(\pi\|\tau)$

Different sources differ slightly in their conventions (for example in whether both players must learn the output, or in the default error probability); such differences change the measures by at most an additive constant or a constant factor, and never matter here. Note that $C(f)$, $C^1(f)$, $C^0(f)$, $C^D(f)$ and $C^P(f)$ always carry the argument (f); a plain C elsewhere denotes a code or a clique.

Standard functions

The four running examples EQ_n (equality), GT_n (greater-than), DISJ_n (disjointness) and IP_n (inner product mod 2) on $\{0,1\}^n \times \{0,1\}^n$ are defined in Definition 1.3. The subscript n is dropped when it is clear from the context.

Other symbols

M_f	communication matrix of f, $M_f[x, y] = f(x, y)$
$\operatorname{rank}(M)$, $\operatorname{rank}_+(M)$	real rank and nonnegative rank
$L(T)$	set of leaves of a protocol tree T
R_v	the rectangle of inputs that reach node v of a protocol tree
f^t, $f^{\wedge t}$, $f^{\vee t}$	t independent copies of f; their AND; their OR
$H(\pi)$, $H(X)$	Shannon entropy of a distribution / random variable
$D_{\mathrm{KL}}(\pi \,\|\, \tau)$	Kullback–Leibler divergence
$I(X;Y)$, $I(X;Y \mid Z)$	mutual information, conditional mutual information
$\mathrm{JS}(\pi, \tau)$	Jensen–Shannon divergence
$h(\pi, \tau)$	Hellinger distance
$\|\pi - \tau\|_1$	ℓ_1 distance $\sum_\omega \|\pi(\omega) - \tau(\omega)\|$ (twice the total variation distance)

Part I

Deterministic Communication

1 The Two-Party Communication Model

The communication model was introduced by Yao [Yao79]. In this chapter we define it informally, record the trivial upper bound, and then look at two problems for which clever protocols beat the trivial bound by an exponential factor. These examples should be kept in mind throughout Part I: most of what follows is about proving that, for other problems, no such clever protocol exists.

1.1 Communication games

Let X, Y, Z be finite sets and let $f\colon X \times Y \to Z$ be a function. Alice receives $x \in X$ and Bob receives $y \in Y$. Neither knows the other's input, and they want to determine $f(x, y)$. To do so they follow a *protocol* agreed upon in advance: they speak in turns, each message being a single bit; who speaks next and what that player sends are determined by the messages exchanged so far together with the speaker's own input. The protocol ends when both players know $f(x, y)$.

Two features of the model deserve emphasis.

- *Computation is free.* Alice and Bob may perform arbitrarily complex local computations. Only the number of exchanged bits is charged.
- *Worst-case cost.* The cost of a protocol is the maximum, over all inputs (x, y), of the number of bits exchanged on that input.

Definition 1.1 (Deterministic communication complexity). The *deterministic communication complexity* of f, denoted $\mathrm{D}(f)$, is the minimum cost of a deterministic protocol computing f.

A formal definition, in terms of *protocol trees*, is given in Section 2.1. For the moment the informal description suffices.

Proposition 1.2 (The trivial protocol). *For every $f\colon X \times Y \to Z$,*

$$\mathrm{D}(f) \;\le\; \lceil \log |X| \rceil + \lceil \log |Z| \rceil .$$

Proof. Alice encodes x by a binary string of length $\lceil \log |X| \rceil$ and sends it. Bob now knows both inputs, computes $f(x, y)$ and sends it back using $\lceil \log |Z| \rceil$ bits. □

By symmetry the same holds with $|Y|$ in place of $|X|$. For Boolean functions on n-bit inputs the trivial protocol costs $n + 1$ bits.

Four Boolean functions on $\{0,1\}^n \times \{0,1\}^n$ will serve as running examples throughout the notes.

Definition 1.3 (Standard functions). For $x, y \in \{0,1\}^n$ (viewed as integers, or as subsets of $[n]$, as convenient):

- *Equality:* $\mathrm{EQ}_n(x, y) = 1$ iff $x = y$;
- *Greater-than:* $\mathrm{GT}_n(x, y) = 1$ iff $x > y$ as integers;
- *Disjointness:* $\mathrm{DISJ}_n(x, y) = 1$ iff $x \cap y = \varnothing$, i.e. there is no i with $x_i = y_i = 1$;
- *Inner product:* $\mathrm{IP}_n(x, y) = \sum_{i=1}^{n} x_i y_i \bmod 2$.

All four have $\mathrm{D} \le n + 1$ by Proposition 1.2. We will see in Chapters 2 and 4 that this is tight (up to an additive constant) for every one of them, whereas in Part II randomness will separate them sharply.

1.2 Example: the median

Alice holds a multiset x of elements of $[n]$ and Bob holds a multiset y of elements of $[n]$; the sizes satisfy $|x|, |y| \le n^c$ for a constant c. They want to compute the median of the multiset union $x \uplus y$.

Convention. For a multiset S of size s, the *median* $\mathrm{MED}(S)$ is its $\lceil s/2 \rceil$-th smallest element.

The trivial protocol costs $O(n^c \log n)$ bits. We will see that $\Theta(\log n)$ bits suffice.

1.2.1 Reduction to the balanced case

First Alice and Bob exchange $|x|$ and $|y|$ ($O(\log n)$ bits). Let $s = |x| + |y|$, and let $N = 2^k$ be the smallest power of two with $N \ge \max(|x|, |y|)$. Both players pad their multisets to size exactly N by adding copies of the smallest possible value 1 and the largest possible value n, in such a way that $t := N - \lceil s/2 \rceil$ copies of 1 and $u := N - \lfloor s/2 \rfloor$ copies of n are added in total (note $t + u = 2N - s$, and how the copies are split between the players is fixed by a rule known to both). The original median had rank $\lceil s/2 \rceil$; after adding t copies of the minimum value, the element of rank $\lceil s/2 \rceil + t = N$ of the padded union has the same value. So from now on we may assume

$$|x| = |y| = N = 2^k, \qquad \mathrm{MED}(x \uplus y) = \text{the } N\text{-th smallest element of } x \uplus y.$$

1.2.2 The halving lemma

Lemma 1.4 (Halving). *Let $|x| = |y| = N = 2^k$ with $k \ge 1$. Let a be the $(N/2)$-th smallest element of x and b the $(N/2)$-th smallest element of y, and suppose $a \le b$. Let x' be obtained from x by deleting its $N/2$ smallest elements and y' be obtained from y by deleting its $N/2$ largest elements. Then*

$$\mathrm{MED}(x \uplus y) = \mathrm{MED}(x' \uplus y'), \qquad \textit{and moreover} \qquad a \le \mathrm{MED}(x \uplus y) \le b.$$

The case $a > b$ is symmetric (exchange the roles of Alice and Bob).

Proof. Write $x = \{a_1 \le \cdots \le a_N\}$ and $y = \{b_1 \le \cdots \le b_N\}$, so that $a = a_{N/2}$ and $b = b_{N/2}$. Sort the $2N$ elements of $x \uplus y$ by value, breaking ties by putting elements of x before elements of y (and keeping the given order inside x and inside y). Let $m^\star$ be the element of rank N in this order; its value is $\mathrm{MED}(x \uplus y)$.

Deleted elements of x lie strictly below rank N. Fix $j \le N/2$. The elements preceding a_j are $a_1, \ldots, a_{j-1}$ and the elements of y of value strictly smaller than a_j. Since $a_j \le a \le b = b_{N/2}$, each of $b_{N/2}, \ldots, b_N$ has value at least a_j, so at most $N/2 - 1$ elements of y precede a_j. Hence the rank of a_j is at most $(j-1) + (N/2 - 1) + 1 \le N - 1$.

Deleted elements of y lie strictly above rank N. Fix $j \ge N/2 + 1$. The element b_j is preceded by $b_1, \ldots, b_{j-1}$ (at least $N/2$ elements) and by every a_l with $l \le N/2$ (as $a_l \le a \le b \le b_j$ and ties put x first). Hence its rank is at least $N/2 + N/2 + 1 = N + 1$.

We have thus removed $N/2$ elements of rank $< N$ and $N/2$ elements of rank $> N$. The element $m^\star$ survives and now has rank $N - N/2 = N/2$ among the remaining N elements, which is exactly the median rank $\lceil N/2 \rceil$ of $x' \uplus y'$. This proves the first claim.

For the second claim: $a = a_{N/2}$ has rank at most $N - 1$ (shown above), so $a \le m^\star$; and $b = b_{N/2}$ is preceded by $b_1, \ldots, b_{N/2-1}$ and by $a_1, \ldots, a_{N/2}$, so its rank is at least N and $m^\star \le b$. □

1.2.3 Two protocols

Protocol 1.5 (Median, first version). Repeat while $N \geq 2$: Alice sends a (the $(N/2)$-th smallest element of her current multiset) and Bob sends b, each using $\lceil \log n \rceil$ bits. Both now know whether $a \leq b$ and update as in Lemma 1.4; N is halved. When $N = 1$, Alice and Bob exchange their single remaining elements and output the smaller one.

There are $k = O(\log n)$ rounds of $O(\log n)$ bits each, so Protocol 1.5 costs $O(\log^2 n)$.

The second version saves a logarithmic factor by observing that in each round the players only need to *compare* a and b, and that comparisons in later rounds can reuse what earlier rounds revealed.

Protocol 1.6 (Median, second version). Write every value as a binary string of length $L = \lceil \log(n+1) \rceil$. Both players maintain a common binary string P (initially empty) with the invariant

$$(\star) \qquad \text{the binary representation of } \mathrm{MED}(x \uplus y) \text{ begins with } P.$$

Let I_P be the set of values whose representation begins with P; this is an interval. In a round with current medians a (Alice) and b (Bob):

1. Each player announces, with two bits, whether its median lies below I_P, inside I_P, or above I_P.
2. If at least one of a, b lies outside I_P, the comparison of a and b is determined (see below) and P is unchanged.
3. If both lie inside I_P, they compare a and b bit by bit, starting at position $|P| + 1$: Alice sends her next bit and Bob answers whether his bit is the same. At the first position q where they differ, the comparison is known; P is replaced by the common prefix of a and b, of length $q - 1$. (If they never differ, then $a = b$ and a is the median by Lemma 1.4; stop.)
4. Both players halve their multisets according to Lemma 1.4.

When $N = 1$ the median is the smaller of the two remaining elements; they compare them in the same way, after which the owner of the smaller one sends its remaining $L - |P|$ bits.

Theorem 1.7. $\mathrm{D}(\mathrm{MED}) = \Theta(\log n)$.

Proof. Correctness of Protocol 1.6. By Lemma 1.4 the median $m^\star$ always lies between a and b. In step 2, a and b cannot both lie above I_P (nor both below), because $m^\star \in I_P$ lies between them; so if, say, a lies above I_P then necessarily $b \leq m^\star < a$, and the comparison is known. In step 3, a and b agree on their first $q - 1$ bits; every integer between a and b then agrees with them on these bits, and since $m^\star$ lies between a and b, the invariant $(\star)$ holds for the longer prefix.

Cost. Steps 1, 2 and 4 cost $O(1)$ bits per round, and there are $k = O(\log n)$ rounds. In step 3 the players spend 2 bits per examined position, and every examined position except the last one is appended to P. Since $|P| \leq L$ at all times, the total cost of all executions of step 3 is at most $2L + 2k$. The final step costs $O(L)$. Together with the $O(\log n)$ bits for the padding phase, the total is $O(\log n)$.

Lower bound. Every value $v \in [n]$ is the median of some input (take $x = y = \{v\}$). A protocol whose transcripts determine the output must therefore have at least n distinct transcripts, i.e. cost at least $\log n$. (This is the range bound of Proposition 2.2.) □

1.3 Example: clique versus independent set

Fix a graph $G = (V, E)$ on n vertices, known to both players. Alice receives a clique $C \subseteq V$ and Bob an independent set $I \subseteq V$. A clique and an independent set share at most one vertex, so

$$\mathrm{CIS}_G(C, I) := |C \cap I| \in \{0, 1\}.$$

The trivial protocol costs about n bits. Yannakakis [Yan91] found a much better one.

For $U \subseteq V$ and $v \in U$ write $\deg_U(v)$ for the number of neighbours of v inside U.

Protocol 1.8 (Clique vs. independent set). Both players maintain a set $U \subseteq V$, initially $U = V$, with the invariant $C \cap I \subseteq U$. A round proceeds as follows.

1. If $U = \varnothing$, output 0.
2. Alice looks for $u \in C \cap U$ with $\deg_U(u) < |U|/2$. If she finds one, she sends it ($\lceil \log n \rceil + 1$ bits, including a flag). Bob replies whether $u \in I$; if so, output 1. Otherwise both set $U \leftarrow U \cap N(u)$.
3. Otherwise Alice sends a failure flag. Bob looks for $w \in I \cap U$ with $\deg_U(w) \geq |U|/2$. If he finds one he sends it; Alice replies whether $w \in C$; if so, output 1. Otherwise both set $U \leftarrow U \setminus (N(w) \cup \{w\})$.
4. If Bob also fails, output 0.

Theorem 1.9 (Yannakakis [Yan91]). *For every graph G on n vertices, $\mathrm{D}(\mathrm{CIS}_G) = O(\log^2 n)$.*

Proof. The invariant is preserved. In step 2, if $u \notin I$ then any vertex of $C \cap I$ is a vertex of C other than u, hence adjacent to u because C is a clique. In step 3, if $w \notin C$ then any vertex of $C \cap I$ is a vertex of I other than w, hence *not* adjacent to w because I is independent.

The output 0 in step 4 is correct. If both players fail, every vertex of $C \cap U$ has $\deg_U \geq |U|/2$ and every vertex of $I \cap U$ has $\deg_U < |U|/2$, so no vertex lies in $C \cap I \cap U$; by the invariant, $C \cap I = \varnothing$.

Cost. In step 2 the new set has size $\deg_U(u) < |U|/2$; in step 3 it has size $|U| - \deg_U(w) - 1 < |U|/2$. Hence U is at least halved in every round that does not stop, so there are at most $\log n + 1$ rounds, each costing $O(\log n)$ bits. □

Remark 1.10. The rank method of Chapter 4 only shows $\mathrm{D}(\mathrm{CIS}_G) \geq \Omega(\log n)$ for suitable G (see Example 4.7). For a long time it was open whether $O(\log n)$ suffices for every G. Göös, Pitassi and Watson [GPW15] showed that this is false: there are graphs with $\mathrm{D}(\mathrm{CIS}_G) \geq \widetilde{\Omega}(\log^2 n)$, so Theorem 1.9 is essentially optimal. As we shall see in Section 4.5, this also settles that the exponent in the log-rank conjecture would have to be at least 2. Yannakakis's original motivation came from the theory of linear programs and extended formulations; see [Yan91] and [RY20, Ch. 12].

Looking ahead. Both examples beat the trivial protocol by an exponential factor, and in both cases the cleverness consisted in making every message cut the remaining "search space" in half. Is there always such a protocol? For equality, disjointness and inner product the answer is no: they need essentially n bits. Proving statements of this kind requires understanding the *structure* that every protocol must have, which is the subject of the next chapter.

1.4 Notes and references

The two-party model is due to Yao [Yao79]. Standard references are the books of Kushilevitz and Nisan [KN97, Ch. 1] and of Rao and Yehudayoff [RY20, Ch. 1]; the survey of Lee and Shraibman [LS09] concentrates on lower-bound techniques. The median problem is a classical

warm-up exercise (see [KN97, Ch. 1]); the clique versus independent set problem and Theorem 1.9 are from Yannakakis [Yan91], and the matching lower bound $\widetilde{\Omega}(\log^2 n)$ is from Göös–Pitassi–Watson [GPW15].

2 Protocol Trees and Combinatorial Rectangles

Much of communication complexity is about *lower bounds*. To prove that no protocol can be cheap, we first need a precise description of what a protocol is. This chapter formalises protocols as trees, shows that the inputs reaching any node of the tree form a *combinatorial rectangle*, and derives the basic lower-bound methods from this single fact. In the second half we go in the opposite direction and show that partitions into rectangles can be turned back into protocols, with only a moderate loss.

2.1 Protocol trees

Definition 2.1 (Protocol tree). A (deterministic) *protocol tree* for inputs in $X \times Y$ with outputs in Z is a rooted binary tree T such that

- every internal node v is *owned* either by Alice or by Bob; an Alice node is labelled by a function $a_v \colon X \to \{0,1\}$ and a Bob node by a function $b_v \colon Y \to \{0,1\}$;
- every leaf ℓ is labelled by an output $z_\ell \in Z$.

On input (x, y) the protocol starts at the root; at an Alice node v it moves to the left child if $a_v(x) = 0$ and to the right child if $a_v(x) = 1$, and similarly at a Bob node with $b_v(y)$. It stops at a leaf ℓ and outputs z_ℓ; we write $T(x, y) = z_\ell$. The tree *computes* f if $T(x, y) = f(x, y)$ for all (x, y). The *cost* of T is its depth.

The bits sent along the way are exactly the directions taken, so the node reached after k messages encodes the whole history of the conversation. Both players can follow the path (each knows the bits sent so far), so at the end both know the leaf, hence the output. The informal protocols of Chapter 1 are therefore protocol trees, and

$$\mathrm{D}(f) = \min\{\mathrm{depth}(T) : T \text{ computes } f\}.$$

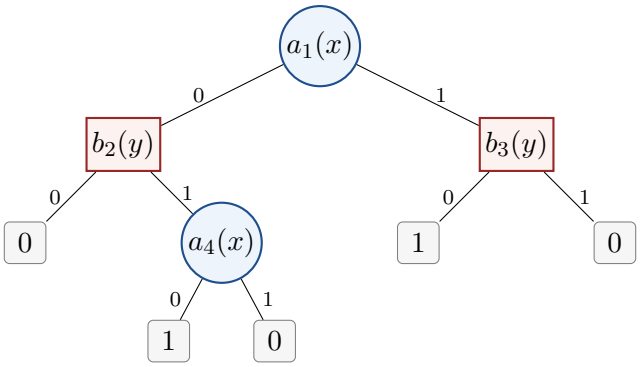


Figure 2.1: A protocol tree of depth 3 with five leaves. Circles are Alice nodes, squares are Bob nodes, and leaves carry outputs.

A binary tree of depth d has at most 2^d leaves, which gives the first (and weakest) lower bound.

Proposition 2.2 (Range bound). *If T computes f and $L(T)$ is its set of leaves, then*

$$\mathrm{depth}(T) \geq \log |L(T)| \geq \log |\,\mathrm{Range}(f)|.$$

In particular $\mathrm{D}(f) \geq \log |\,\mathrm{Range}(f)|$.

Proof. $|L(T)| \le 2^{\mathrm{depth}(T)}$, and every value of f must appear as the label of at least one leaf. □

For Boolean functions the range bound is useless. The next two sections develop bounds that are useful.

2.2 Rectangles

Definition 2.3 (Rectangle). A *(combinatorial) rectangle* in $X \times Y$ is a set of the form $A \times B$ with $A \subseteq X$ and $B \subseteq Y$.

A rectangle need not be contiguous: after permuting rows and columns of the matrix $X \times Y$ it becomes a contiguous block. The following characterisation is often the most convenient one.

Lemma 2.4. *A set $S \subseteq X \times Y$ is a rectangle if and only if*

$$(x, y) \in S \text{ and } (x', y') \in S \quad \Longrightarrow \quad (x, y') \in S.$$

Proof. If $S = A \times B$ and $(x, y), (x', y') \in S$, then $x \in A$ and $y' \in B$, so $(x, y') \in S$. Conversely, let A and B be the projections of S to X and to Y. Clearly $S \subseteq A \times B$. If $x \in A$ and $y' \in B$, there are y, x' with $(x, y), (x', y') \in S$, so $(x, y') \in S$ by the closure property; hence $S = A \times B$. □

Lemma 2.5 (Nodes are rectangles). *For every node v of a protocol tree, the set R_v of inputs (x, y) whose computation passes through v is a rectangle $A_v \times B_v$.*

Proof. By induction on the depth of v. The root has $R_v = X \times Y$. Let v be an Alice node with $R_v = A_v \times B_v$ and children v_0, v_1. An input in R_v goes to v_b iff $a_v(x) = b$, a condition on x alone, so $R_{v_b} = (A_v \cap a_v^{-1}(b)) \times B_v$. Bob nodes are symmetric. □

Definition 2.6 (Partition number). A rectangle R is *f-monochromatic* if f is constant on R; if the constant is z we call it a *z-rectangle*. The *partition number* $C^D(f)$ is the minimum number of monochromatic rectangles in a partition of $X \times Y$.

Proposition 2.7. *If T computes f, the leaf rectangles $\{R_\ell\}_{\ell \in L(T)}$ form a partition of $X \times Y$ into f-monochromatic rectangles. Consequently*

$$\mathrm{D}(f) \;\ge\; \log C^D(f).$$

Proof. Every input reaches exactly one leaf, so the R_ℓ partition $X \times Y$; each R_ℓ is a rectangle by Lemma 2.5, and f equals z_ℓ on R_ℓ because T computes f. So $C^D(f) \le |L(T)| \le 2^{\mathrm{depth}(T)}$. □

Thus every lower bound on $C^D(f)$ is a lower bound on communication. The remaining sections of this chapter, and all of Chapter 4, are about ways to bound $C^D(f)$ from below.

2.3 Fooling sets

Definition 2.8 (Fooling set). A set $S \subseteq X \times Y$ is a *fooling set* for f if there is a value z such that

(i) $f(x, y) = z$ for every $(x, y) \in S$, and

(ii) for any two distinct $(x_1, y_1), (x_2, y_2) \in S$, either $f(x_1, y_2) \neq z$ or $f(x_2, y_1) \neq z$.

Lemma 2.9 (Fooling-set bound). *If S is a fooling set for f with value z, then every partition (and even every covering) of $f^{-1}(z)$ by z-rectangles uses at least $|S|$ rectangles. In particular $C^D(f) \ge |S|$ and $\mathrm{D}(f) \ge \log |S|$.*

Proof. Suppose a z-rectangle R contains two distinct points (x_1, y_1) and (x_2, y_2) of S. By Lemma 2.4, R also contains (x_1, y_2) and (x_2, y_1), so f takes the value z on both of them, contradicting (ii). Hence each z-rectangle contains at most one point of S, and at least $|S|$ of them are needed to cover S. □

Example 2.10 *(Equality).* $S = \{(x, x) : x \in \{0,1\}^n\}$ is a fooling set for EQ_n with $z = 1$: for $x_1 \neq x_2$ we have $\mathrm{EQ}(x_1, x_2) = 0$. So at least 2^n rectangles are needed to partition the 1-entries, and at least one more for the 0-entries. Hence $C^D(\mathrm{EQ}_n) \geq 2^n + 1$ and

$$\mathrm{D}(\mathrm{EQ}_n) \geq \lceil \log(2^n + 1) \rceil = n + 1,$$

which matches the trivial protocol exactly.

Example 2.11 *(Greater-than).* $S = \{(t, t-1) : 1 \leq t \leq 2^n - 1\}$ is a fooling set for GT_n with $z = 1$: for $t < t'$, the pair $(t, t'-1)$ has $t \leq t' - 1$, so $\mathrm{GT}(t, t'-1) = 0$. Since GT also takes the value 0, $C^D(\mathrm{GT}_n) \geq 2^n$ and $\mathrm{D}(\mathrm{GT}_n) \geq n$.

Example 2.12 *(Disjointness).* $S = \{(x, \bar{x}) : x \subseteq [n]\}$, where $\bar{x} = [n] \setminus x$, is a fooling set for DISJ_n with $z = 1$: if $x_1 \neq x_2$, then some element lies in one of them but not the other, say $i \in x_1 \setminus x_2$; then $i \in x_1 \cap \bar{x}_2$, so $\mathrm{DISJ}(x_1, \bar{x}_2) = 0$. As for equality, $\mathrm{D}(\mathrm{DISJ}_n) \geq n + 1$, which is tight.

2.4 The rectangle-size bound

Fooling sets count points that no rectangle can share. A weighted version of the same idea is often more flexible.

Lemma 2.13 (Rectangle-size bound). *Let μ be a probability distribution on $X \times Y$ and suppose that every f-monochromatic rectangle R has $\mu(R) \leq \delta$. Then $C^D(f) \geq 1/\delta$ and hence $\mathrm{D}(f) \geq \log(1/\delta)$.*

Proof. If $X \times Y$ is partitioned into t monochromatic rectangles, then $1 = \sum_{i=1}^{t} \mu(R_i) \leq t\delta$. □

The fooling-set bound is the special case where μ is uniform on the fooling set S: a z-rectangle contains at most one point of S, and a rectangle of any other colour contains none, so $\delta = 1/|S|$. The real strength of Lemma 2.13 is that it can be applied when no large fooling set is apparent, as in the following example.

Example 2.14 *(Inner product).* Let μ be uniform on $\{0,1\}^n \times \{0,1\}^n$. We claim that every IP_n-monochromatic rectangle $A \times B$ satisfies $|A| \cdot |B| \leq 2^n$, so that $\mu(A \times B) \leq 2^{-n}$ and $\mathrm{D}(\mathrm{IP}_n) \geq n$.

Work over $\mathbb{F}_2$. If $A \times B$ is a 0-rectangle, then $\langle a, b \rangle = 0$ for all $a \in A$, $b \in B$, so $\mathrm{span}(B) \subseteq \mathrm{span}(A)^{\perp}$ and $\dim \mathrm{span}(A) + \dim \mathrm{span}(B) \leq n$. Therefore $|A|\,|B| \leq 2^{\dim \mathrm{span}(A)}\, 2^{\dim \mathrm{span}(B)} \leq 2^n$.

If $A \times B$ is a 1-rectangle, fix $a_0 \in A$, $b_0 \in B$ and put $A' = A + a_0$, $B' = B + b_0$. For $a \in A$, $b \in B$,

$$\langle a + a_0, b + b_0 \rangle = \langle a, b \rangle + \langle a, b_0 \rangle + \langle a_0, b \rangle + \langle a_0, b_0 \rangle = 1 + 1 + 1 + 1 = 0,$$

so $A' \times B'$ is "orthogonal" and the previous argument gives $|A|\,|B| = |A'|\,|B'| \leq 2^n$.

2.5 Counting leaves and balancing protocols

The partition number gives lower bounds. Can it also give upper bounds? Not every monochromatic partition comes from a protocol tree, so we first introduce an intermediate measure.

Definition 2.15 (Protocol partition number). $C^P(f)$ is the minimum number of leaves of a protocol tree computing f.

Proposition 2.16. *$C^D(f) \le C^P(f) \le 2^{\mathrm{D}(f)}$; hence $\log C^D(f) \le \log C^P(f) \le \mathrm{D}(f)$.*

Proof. The leaves of an optimal tree form a monochromatic partition of size $C^P(f)$ (Proposition 2.7), and a tree of depth $\mathrm{D}(f)$ has at most $2^{\mathrm{D}(f)}$ leaves. □

Example 2.17 *($C^D < C^P$).* Let $X = Y = \{1,2,3\}$ and partition the 3×3 grid into the five rectangles of Figure 2.2:

$$\{1,2\} \times \{1\}, \quad \{3\} \times \{1,2\}, \quad \{2,3\} \times \{3\}, \quad \{1\} \times \{2,3\}, \quad \{2\} \times \{2\}.$$

Let f assign a different colour to each of the five rectangles, so $C^D(f) = 5$. Any protocol tree for f has monochromatic leaves, so each leaf rectangle lies inside one of the five. If some tree had exactly five leaves, its leaves would be precisely these five rectangles. Consider the first message. If Alice speaks first, she splits X into two nonempty parts with every leaf rectangle on one side; but the row sets $\{1,2\}$ and $\{2,3\}$ force $1,2,3$ to lie on the same side, a contradiction. The column sets $\{1,2\}$ and $\{2,3\}$ rule out Bob in the same way. Hence $C^P(f) > 5 = C^D(f)$.

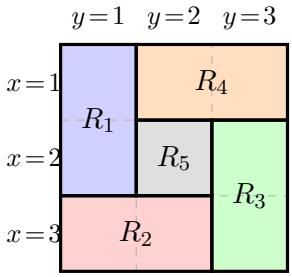


Figure 2.2: A partition of $\{1,2,3\}^2$ into five rectangles that does not arise from any protocol tree: no first message can separate the rectangles.

Although C^D and C^P can differ, C^P determines D up to a constant factor. The key is that every tree can be *balanced*.

Lemma 2.18 (Tree separator). *Let T be a rooted binary tree with $m \ge 2$ leaves. There is a node u whose subtree T_u satisfies $m/3 < |L(T_u)| \le 2m/3$.*

Proof. Start at the root, which has $m > 2m/3$ leaves below it, and repeatedly move to a child with more than $2m/3$ leaves below it, as long as such a child exists. The walk stops at a node w with $|L(T_w)| > 2m/3 \ge 4/3$, so w is internal, and both children of w have at most $2m/3$ leaves. The child u with more leaves has $|L(T_u)| \ge |L(T_w)|/2 > m/3$. □

Theorem 2.19 (Balancing). *For every f,*

$$\log C^P(f) \;\le\; \mathrm{D}(f) \;\le\; 2\left\lceil \log_{3/2} C^P(f) \right\rceil.$$

In particular $\mathrm{D}(f) = \Theta(\log C^P(f))$.

Proof. The lower bound is Proposition 2.16. For the upper bound we prove by induction on m: *for every protocol tree T with m leaves there is a protocol of cost at most $2\left\lceil \log_{3/2} m \right\rceil$ that outputs $T(x,y)$ on every input.* For $m = 1$ the output is fixed and no communication is needed.

Let $m \ge 2$ and choose u as in Lemma 2.18, with $R_u = A_u \times B_u$. Alice sends the bit $[x \in A_u]$ and Bob sends $[y \in B_u]$; now both know whether $(x,y) \in R_u$.

- If $(x,y) \in R_u$, the computation of T passes through u, so $T(x,y) = T_u(x,y)$, where T_u is viewed as a protocol tree in its own right. It has at most $2m/3$ leaves.

- If $(x, y) \notin R_u$, let T' be obtained from T by deleting the subtree T_u and contracting its parent p (the sibling of u takes the place of p). An input that reaches p but not u is sent by p to the sibling, so $T'(x, y) = T(x, y)$ for every $(x, y) \notin R_u$. The tree T' has $m - |L(T_u)| < 2m/3$ leaves.

In both cases the players continue with a tree having at most $\lfloor 2m/3 \rfloor$ leaves. By induction the total cost is at most $2 + 2\left\lceil \log_{3/2}(2m/3) \right\rceil \le 2\left\lceil \log_{3/2} m \right\rceil$. □

2.6 From monochromatic partitions to protocols

Does a small monochromatic partition always give a cheap protocol, even if the partition itself is not a protocol? The answer is yes, with a quadratic loss.

Definition 2.20. Two rectangles $A \times B$ and $A' \times B'$ *intersect horizontally* if $A \cap A' \neq \varnothing$ (they share a row) and *intersect vertically* if $B \cap B' \neq \varnothing$ (they share a column).

Lemma 2.21. *Two disjoint rectangles cannot intersect both horizontally and vertically.*

Proof. If $x \in A \cap A'$ and $y \in B \cap B'$ then $(x, y) \in (A \times B) \cap (A' \times B')$. □

Let $\mathcal{R}$ be a family of pairwise disjoint rectangles and $R \in \mathcal{R}$. Let $h_{\mathcal{R}}(R)$ (resp. $v_{\mathcal{R}}(R)$) be the number of members of $\mathcal{R}$, including R itself, that intersect R horizontally (resp. vertically). By Lemma 2.21, every member other than R is counted at most once in $h_{\mathcal{R}}(R) + v_{\mathcal{R}}(R)$, so

$$h_{\mathcal{R}}(R) + v_{\mathcal{R}}(R) \le |\mathcal{R}| + 1. \tag{2.1}$$

Call R *horizontally good* if $h_{\mathcal{R}}(R) \le \lceil |\mathcal{R}|/2 \rceil$ and *vertically good* if $v_{\mathcal{R}}(R) \le \lceil |\mathcal{R}|/2 \rceil$. By (2.1) every $R \in \mathcal{R}$ is horizontally good or vertically good.

Theorem 2.22 ([AUY83]). *For every f, $\mathrm{D}(f) = O(\log^2 C^D(f))$. More precisely, $\mathrm{D}(f) \le \left\lceil \log C^D(f) \right\rceil \left(\left\lceil \log C^D(f) \right\rceil + 2 \right)$.*

Proof. Fix a monochromatic partition $\mathcal{R}$ of $X \times Y$ with $|\mathcal{R}| = C^D(f)$. The players maintain a common subfamily $\mathcal{R}' \subseteq \mathcal{R}$ that contains the unique rectangle $R^\star = A^\star \times B^\star$ containing (x, y). Initially $\mathcal{R}' = \mathcal{R}$. While $|\mathcal{R}'| \ge 2$ they do the following.

1. Alice looks for a horizontally good (with respect to $\mathcal{R}'$) rectangle $A \times B \in \mathcal{R}'$ with $x \in A$. If she finds one she sends 1 followed by its index; both players replace $\mathcal{R}'$ by the members that intersect $A \times B$ horizontally.
2. Otherwise Alice sends 0; then Bob finds a vertically good rectangle $A \times B \in \mathcal{R}'$ with $y \in B$, sends its index, and both replace $\mathcal{R}'$ by the members that intersect $A \times B$ vertically.

When $|\mathcal{R}'| = 1$ the remaining rectangle is $R^\star$, and its colour is the answer.

The invariant is preserved. In step 1, $R^\star$ contains row x, as does $A \times B$; so $R^\star$ intersects $A \times B$ horizontally and is kept. Step 2 is symmetric.

Bob always succeeds in step 2. $R^\star \in \mathcal{R}'$ is horizontally or vertically good, and $x \in A^\star$, $y \in B^\star$. If Alice failed, then $R^\star$ is not horizontally good, so it is vertically good and Bob can pick it (or another suitable rectangle).

Cost. After each round $|\mathcal{R}'| \le \lceil |\mathcal{R}'|/2 \rceil$, so after $\left\lceil \log C^D(f) \right\rceil$ rounds only one rectangle is left. Each round costs at most $2 + \left\lceil \log C^D(f) \right\rceil$ bits. □

Remark 2.23. Theorem 2.22 together with Proposition 2.16 shows $\log C^D(f) \le \mathrm{D}(f) \le O(\log^2 C^D(f))$; in particular C^D and C^P are polynomially related. Is the square necessary? Göös, Pitassi and Watson [GPW15] gave the first function with $\mathrm{D}(f) \ge \widetilde{\Omega}(\log^{1.5} C^D(f))$, and Ambainis, Kokainis and Kothari [AKK16] improved this to an almost quadratic separation. So Theorem 2.22 is essentially optimal.

Looking ahead. The proof of Theorem 2.22 used only one property of the partition: two of its rectangles cannot share both a row and a column unless they are the same rectangle. In the next chapter we drop the requirement that the rectangles be disjoint, which leads to *covers* and to *nondeterministic* protocols, and we will see that the same argument still goes through when rectangles of different colours are involved.

2.7 Notes and references

Protocol trees and the rectangle property go back to Yao [Yao79]; see [KN97, Ch. 1] and [RY20, Ch. 1] for textbook treatments, including the balancing lemma (Theorem 2.19) and the pinwheel example (Example 2.17). The fooling-set method is also from [Yao79]. The rectangle-size bound with the uniform distribution is the basis of the discrepancy method of Chapter 7. Theorem 2.22 is a special case of the theorem of Aho, Ullman and Yannakakis [AUY83] (Theorem 3.8). For the separations between D and $\log C^D$, see [GPW15, AKK16].

3 Nondeterministic Communication

In Chapter 2 we saw that deterministic protocols correspond, up to a square, to *partitions* of the communication matrix into monochromatic rectangles. In this chapter we relax the partition to a *cover*, in which rectangles may overlap. Covers have a natural computational meaning as well.

For some problems one of the two answers has a short *certificate*. If two sets intersect, naming a common element proves it with $\log n$ bits. If they are disjoint, no short proof is apparent. Nondeterministic communication complexity measures the length of such certificates, and it corresponds exactly to covers. The main result of the chapter is that if *both* answers have short certificates, then a short deterministic protocol exists.

3.1 Rectangle covers

Definition 3.1 (Covers). Let $f\colon X \times Y \to \{0,1\}$. A *$z$-cover* of f is a family of z-rectangles whose union is $f^{-1}(z)$. The minimum size of a z-cover is denoted $C^z(f)$, and $C(f)$ denotes the minimum size of a family of monochromatic rectangles whose union is $X \times Y$; thus $C(f) = C^0(f) + C^1(f)$.

Every monochromatic partition is a cover, so $C(f) \le C^D(f)$, and every lower bound on $C^z(f)$ is a lower bound on $C^D(f)$. The fooling-set argument (Lemma 2.9) already gives such lower bounds: a fooling set for the value z forces $C^z(f) \ge |S|$. For example, Examples 2.10 and 2.12 show $C^1(\mathrm{EQ}_n) \ge 2^n$ and $C^1(\mathrm{DISJ}_n) \ge 2^n$.

3.2 Nondeterministic protocols

Definition 3.2 (Nondeterministic protocols). Let $f\colon X \times Y \to \{0,1\}$. A *nondeterministic protocol for the 1-inputs of f* consists of a set of *proofs* $W = \{0,1\}^w$ and, for every $u \in W$, a deterministic protocol P_u with outputs in $\{0,1\}$, such that for every input (x,y):

- if $f(x,y) = 1$, then $P_u(x,y) = 1$ for *some* $u \in W$;
- if $f(x,y) = 0$, then $P_u(x,y) = 0$ for *all* $u \in W$.

Both players see the proof u; its length is charged to the protocol, whose cost is $w + \max_u \mathrm{cost}(P_u)$. The minimum cost is the *nondeterministic complexity* $\mathrm{N}^1(f)$. We set $\mathrm{N}^0(f) := \mathrm{N}^1(1-f)$.

The proof u can be thought of as given by an all-knowing prover, or as a guess that the players verify. Equivalently, a nondeterministic protocol can be described as a protocol tree in which a player may choose among several moves, an input being accepted if *some* sequence of choices leads to an accepting leaf; the sequence of choices is then the proof.

Lemma 3.3 (Accepting leaves are monochromatic rectangles). *A nondeterministic protocol of cost c for the 1-inputs of f yields a 1-cover of f of size at most 2^c.*

Proof. Fix a proof u. Each leaf of the deterministic protocol P_u corresponds to a rectangle (Lemma 2.5). If a leaf with output 1 contained a 0-input, that input would be accepted, which is forbidden. So the accepting leaves of P_u are 1-rectangles. Every 1-input is accepted by some P_u, i.e. lies in an accepting leaf of some P_u. Hence the accepting leaves of all the P_u together cover $f^{-1}(1)$. There are at most $2^w \cdot 2^{\max_u \mathrm{cost}(P_u)} = 2^c$ pairs (u, leaf). □

Remark 3.4 *(Covers versus partitions).* Within one fixed deterministic protocol P_u the leaves are disjoint, and each leaf is exactly one rectangle. The rectangles coming from *different* proofs

may overlap, and an input may be accepted along several paths. This is why nondeterminism corresponds to *covers* rather than *partitions*.

Theorem 3.5. *For every Boolean f and $z \in \{0,1\}$,*

$$\log C^z(f) \;\le\; \mathrm{N}^z(f) \;\le\; \lceil \log C^z(f) \rceil + 2.$$

Proof. The lower bound is Lemma 3.3 (applied to f or $1-f$). For the upper bound, fix a z-cover $A_1 \times B_1, \dots, A_k \times B_k$ with $k = C^z(f)$. The proof is an index $i \in [k]$; given it, Alice sends $[x \in A_i]$ and Bob sends $[y \in B_i]$, and they accept iff both bits are 1. An input is accepted under proof i iff it lies in $A_i \times B_i$; so the z-inputs are exactly the inputs accepted under some proof. □

Example 3.6 *(Equality).* $\mathrm{N}^0(\mathrm{EQ}_n) \le \lceil \log n \rceil + 2$: the proof is an index i with $x_i \neq y_i$; Alice sends x_i and Bob checks it. On the other hand $C^1(\mathrm{EQ}_n) \ge 2^n$, so $\mathrm{N}^1(\mathrm{EQ}_n) \ge n$.

Example 3.7 *(Disjointness).* $\mathrm{N}^0(\mathrm{DISJ}_n) \le \lceil \log n \rceil + 2$: the proof is a common element i of x and y. Conversely, $\{(\{i\},\{i\}) : i \in [n]\}$ is a fooling set for the value 0 (the pair $(\{i\},\{j\})$ with $i \neq j$ is disjoint), so $C^0(\mathrm{DISJ}_n) \ge n$ and $\mathrm{N}^0(\mathrm{DISJ}_n) \ge \log n$. The fooling set of Example 2.12 gives $C^1(\mathrm{DISJ}_n) \ge 2^n$ and $\mathrm{N}^1(\mathrm{DISJ}_n) \ge n$.

So nondeterministic complexity can be exponentially smaller than deterministic complexity, but, as these examples suggest, not on both sides simultaneously.

3.3 Certificates on both sides

The key observation is the same as in Section 2.6: a 1-rectangle and a 0-rectangle are disjoint, so by Lemma 2.21 they cannot share both a row and a column.

Theorem 3.8 (Aho–Ullman–Yannakakis [AUY83]). *For every Boolean f,*

$$\mathrm{D}(f) = O\big(\log C^0(f) \cdot \log C^1(f)\big) = O\big(\mathrm{N}^0(f)\,\mathrm{N}^1(f)\big),$$

(assuming $C^0(f), C^1(f) \ge 2$; otherwise f is constant or the statement is trivial).

Proof. Fix a 1-cover $\{R_i = A_i \times B_i\}_{i \in [a]}$ and a 0-cover $\{S_j = U_j \times V_j\}_{j \in [b]}$ with $a = C^1(f)$ and $b = C^0(f)$. The players maintain a common set $\mathcal{S} \subseteq [b]$ of 0-rectangles with the invariant

$$(\star) \qquad f(x,y) = 0 \implies (x,y) \in S_j \text{ for some } j \in \mathcal{S}.$$

Initially $\mathcal{S} = [b]$. While $\mathcal{S} \neq \varnothing$:

1. Alice looks for an i with $x \in A_i$ such that R_i intersects horizontally at most half of the rectangles S_j, $j \in \mathcal{S}$. If she finds one, she sends i, and both players replace $\mathcal{S}$ by $\{j \in \mathcal{S} : S_j \text{ intersects } R_i \text{ horizontally}\}$.
2. Otherwise Alice says so, and Bob looks for an i with $y \in B_i$ such that R_i intersects vertically at most half of the S_j, $j \in \mathcal{S}$. If he finds one, he sends i and both replace $\mathcal{S}$ by the j with S_j intersecting R_i vertically.
3. If neither finds such an i, output 0.

When $\mathcal{S} = \varnothing$, output 1.

The invariant is preserved. In step 1, if $f(x,y) = 0$ and $(x,y) \in S_j$, then S_j contains row x, as does R_i; so S_j intersects R_i horizontally and j is kept. Step 2 is symmetric.

Output 1 *is correct.* If $\mathcal{S} = \varnothing$, the invariant says $f(x,y) \neq 0$.

Output 0 *is correct.* Suppose $f(x,y) = 1$, say $(x,y) \in R_i$. Each S_j intersects R_i horizontally, vertically, or neither, but never both (Lemma 2.21). So either at most half of the S_j ($j \in \mathcal{S}$)

intersect R_i horizontally, and then Alice can choose i (as $x \in A_i$), or at most half intersect it vertically, and then Bob can choose i (as $y \in B_i$). So if both players fail, $f(x, y) = 0$.

Cost. Each successful round at least halves $\mathcal{S}$, so there are at most $\log b + 1$ rounds, each costing $O(\log a)$ bits. □

Remark 3.9. Theorem 3.8 implies Theorem 2.22, because $C^0(f), C^1(f) \le C^D(f)$. It also gives, for later use,

$$\mathrm{D}(f) = O\big(\log^2 C(f)\big), \tag{3.1}$$

where $C(f) = C^0(f) + C^1(f)$ is the minimum size of a cover of the whole matrix by monochromatic rectangles. In the next chapter we will meet a strengthening due to Lovász (Theorem 4.12), in which $\log C^0$ is replaced by the logarithm of the rank of the communication matrix.

Remark 3.10. For particular functions the bound can be loose: $\mathrm{D}(\mathrm{EQ}_n) = n + 1$, whereas $\mathrm{N}^0(\mathrm{EQ}_n)\,\mathrm{N}^1(\mathrm{EQ}_n) = \Theta(n \log n)$. Whether, and how far, the product $\mathrm{N}^0\mathrm{N}^1$ can be approached by D is closely tied to the separations between D and $\log C^D$ mentioned after Theorem 2.22; see [KN97, Ch. 2] and [GPW15, AKK16].

Looking ahead. All lower bounds so far were obtained by exhibiting, by hand, a set of inputs that no rectangle can cover efficiently (fooling sets and the rectangle-size bound). The next chapter introduces a systematic algebraic tool: since a monochromatic rectangle is a matrix of rank at most one, the rank of the communication matrix bounds the number of rectangles from below. Rank will also give new *upper* bounds, and covers will reappear there.

3.4 Notes and references

Nondeterministic communication and its relation to covers are treated in [KN97, Ch. 2] and [RY20, Ch. 1]. Theorem 3.8 is due to Aho, Ullman and Yannakakis [AUY83]; Lovász [Lov90] gave the refinement in Theorem 4.12. The connection between nondeterminism, covers and the clique versus independent set problem is discussed in [Yan91, GPW15].

4 Rank and the Log-Rank Conjecture

The lower bounds of the previous two chapters count rectangles by combinatorial arguments that have to be invented anew for every function. In this chapter we develop an algebraic method that is often much easier to apply. From now on f is usually Boolean, and we study it through its *communication matrix*

$$M_f \in \{0,1\}^{X\times Y}, \qquad M_f[x,y] = f(x,y).$$

A monochromatic rectangle of colour 1 is an all-ones submatrix, which is a matrix of rank one. This simple observation connects communication complexity with linear algebra, and leads both to lower bounds (rank is easy to compute) and to upper bounds. It also leads to the most famous open problem of the field, the log-rank conjecture.

4.1 Rank, nonnegative rank and the rank lower bound

Recall that the rank of an $m \times n$ real matrix M is the smallest r such that $M = AB$ with $A \in \mathbb{R}^{m\times r}$ and $B \in \mathbb{R}^{r\times n}$; equivalently, the smallest number of rank-one matrices summing to M. Requiring the factors to be nonnegative gives a different notion.

Definition 4.1 (Nonnegative rank). For a matrix M with nonnegative entries, $\operatorname{rank}_+(M)$ is the smallest r such that $M = AB$ with $A \in \mathbb{R}^{m\times r}_{\geq 0}$ and $B \in \mathbb{R}^{r\times n}_{\geq 0}$; equivalently, the smallest number of nonnegative rank-one matrices summing to M.

Clearly $\operatorname{rank}(M) \leq \operatorname{rank}_+(M) \leq \min(m,n)$.

Remark 4.2. The two notions can be very far apart. For instance, the *unique-disjointness* matrix, indexed by $a, b \subseteq [n]$ with entries $(1-|a\cap b|)^2$, has rank at most $(n+1)^2$ but nonnegative rank $2^{\Omega(n)}$; this is the heart of the lower bounds on extended formulations of Fiorini et al. [FMP+15]. Moreover, rank is computable by Gaussian elimination, while computing rank_+ is NP-hard [Vav09].

Theorem 4.3 (Rank lower bound [MS82]). *For every Boolean f,*

$$\log \operatorname{rank}(M_f) \;\leq\; \log \operatorname{rank}_+(M_f) \;\leq\; \log C^D(f) \;\leq\; \mathrm{D}(f).$$

Proof. The first inequality is immediate and the last is Proposition 2.7. For the middle one, let $R_1, \dots, R_t$ be a monochromatic partition with $t = C^D(f)$, $R_i = A_i \times B_i$. Summing the indicator matrices of the 1-rectangles gives

$$M_f = \sum_{i:\ R_i \text{ is a 1-rectangle}} \mathbf{1}_{A_i}\mathbf{1}_{B_i}^{\top},$$

a sum of at most t nonnegative rank-one matrices. □

4.2 Examples

We use two facts from linear algebra: for the Kronecker product, $\operatorname{rank}(A\otimes B) = \operatorname{rank}(A)\operatorname{rank}(B)$; and rank is subadditive, $\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$.

Example 4.4 *(Equality and greater-than).* M_{EQ_n} is the $2^n \times 2^n$ identity matrix, so rank $= 2^n$ and $\mathrm{D}(\mathrm{EQ}_n) \geq n$. With rows and columns ordered by the integer value, M_{GT_n} is strictly lower triangular with all entries below the diagonal equal to 1; its first row is zero and the remaining rows are linearly independent, so $\operatorname{rank}(M_{\mathrm{GT}_n}) = 2^n - 1$.

Example 4.5 *(Disjointness).* Order the subsets of $[n]$ so that the first coordinate splits the rows (and columns) into two halves. The block where $x_1 = y_1 = 1$ is all zero (the sets intersect), and every other block is $M_{\mathrm{DISJ}_{n-1}}$. Hence

$$M_{\mathrm{DISJ}_n} = \begin{pmatrix} M_{\mathrm{DISJ}_{n-1}} & M_{\mathrm{DISJ}_{n-1}} \\ M_{\mathrm{DISJ}_{n-1}} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \otimes M_{\mathrm{DISJ}_{n-1}} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{\otimes n}.$$

The 2×2 matrix has rank 2, so $\mathrm{rank}(M_{\mathrm{DISJ}_n}) = 2^n$.

Example 4.6 *(Inner product).* Consider the sign matrix $H_n[x, y] = (-1)^{\mathrm{IP}(x,y)}$. Since $(-1)^{\mathrm{IP}(x,y)} = \prod_i (-1)^{x_i y_i}$, we get

$$H_n = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{\otimes n},$$

the Sylvester–Hadamard matrix, of rank 2^n. As $H_n = J - 2M_{\mathrm{IP}_n}$ with J the all-ones matrix, $2^n = \mathrm{rank}(H_n) \le 1 + \mathrm{rank}(M_{\mathrm{IP}_n})$, so $\mathrm{rank}(M_{\mathrm{IP}_n}) \ge 2^n - 1$ and $\mathrm{D}(\mathrm{IP}_n) \ge \log(2^n - 1)$.

Example 4.7 *(Clique vs. independent set).* For $v \in V$ let a_v be the vector indexed by cliques with $a_v(C) = [v \in C]$ and b_v the vector indexed by independent sets with $b_v(I) = [v \in I]$. Then

$$M_{\mathrm{CIS}_G}[C, I] = |C \cap I| = \sum_{v \in V} a_v(C)\, b_v(I), \qquad \text{i.e.} \qquad M_{\mathrm{CIS}_G} = \sum_{v \in V} a_v b_v^{\top},$$

a sum of n nonnegative rank-one matrices. So both $\mathrm{rank}(M_{\mathrm{CIS}_G})$ and $\mathrm{rank}_+(M_{\mathrm{CIS}_G})$ are at most n. Thus the rank method yields at most $\log n$ for CIS_G, while Theorem 1.9 gives $O(\log^2 n)$. (For the edgeless graph the cliques are $\varnothing$ and the singletons, and $\{(\{v\}, \{v\}) : v \in V\}$ is a fooling set, so $\mathrm{D}(\mathrm{CIS}_G) \ge \log n$ for that G.)

4.3 Upper bounds from rank

Lemma 4.8. *A Boolean matrix of rank r has at most 2^r distinct rows and at most 2^r distinct columns.*

Proof. Choose a set K of r columns forming a basis of the column space. Every column c is a fixed linear combination $c = \sum_{k \in K} \alpha_{c,k}\, k$ of the columns in K, so every entry of row x is determined by the entries of row x in the columns of K: $M[x, c] = \sum_{k \in K} \alpha_{c,k} M[x, k]$. Hence the restriction of a row to K determines the row. There are at most 2^r Boolean vectors of length r. Columns are handled symmetrically. □

Proposition 4.9. *For Boolean f, $\mathrm{D}(f) \le \mathrm{rank}(M_f) + 1$.*

Proof. With K as in the previous proof, Alice sends the r bits $(M_f[x, k])_{k \in K}$. Bob computes $M_f[x, y] = \sum_k \alpha_{y,k} M_f[x, k]$ and sends back the answer. □

So $\log \mathrm{rank}(M_f) \le \mathrm{D}(f) \le \mathrm{rank}(M_f) + 1$: an exponential gap. Where does the truth lie? Before discussing this question (Section 4.5), we prove a bound that combines rank with *covers*.

4.4 Covers and Lovász's bound

Recall from Definition 3.1 that $C^1(f)$ is the minimum number of 1-rectangles needed to cover all 1-entries of M_f, and that $\log C^1(f)$ is, up to an additive constant, the nondeterministic complexity $\mathrm{N}^1(f)$ (Theorem 3.5). Nonnegative rank is an upper bound for it.

Lemma 4.10. *For Boolean f, $C^1(f) \le \mathrm{rank}_+(M_f)$.*

Proof. Write $M_f = \sum_{i=1}^{r} u_i v_i^{\top}$ with $r = \operatorname{rank}_+(M_f)$ and u_i, v_i nonnegative vectors (we may assume $u_i, v_i \neq 0$). The support of $u_i v_i^{\top}$ is the rectangle $\operatorname{supp}(u_i) \times \operatorname{supp}(v_i)$. Since all summands are nonnegative, every entry in this support is positive in M_f, i.e. equal to 1; so it is a 1-rectangle. Every 1-entry of M_f is positive in at least one summand, so these r rectangles form a 1-cover. □

The next lemma is the linear-algebra heart of the argument. It says that cutting a matrix along a rank-one block splits the rank roughly in two.

Lemma 4.11 (Rank splitting). *Let* $M = \begin{pmatrix} N & U \\ L & D \end{pmatrix}$ *be a block matrix. Then*

$$\operatorname{rank}\begin{pmatrix} N & U \end{pmatrix} + \operatorname{rank}\begin{pmatrix} N \\ L \end{pmatrix} \;\leq\; \operatorname{rank}(M) + 3\operatorname{rank}(N).$$

In particular, if N is a monochromatic block of a Boolean matrix (so $\operatorname{rank} N \leq 1$), the left-hand side is at most $\operatorname{rank}(M) + 3$.

Proof. By subadditivity, $\operatorname{rank}(N\ U) \leq \operatorname{rank}(N) + \operatorname{rank}(U)$ and $\operatorname{rank}\binom{N}{L} \leq \operatorname{rank}(N) + \operatorname{rank}(L)$. Next,

$$\operatorname{rank}(U) + \operatorname{rank}(L) \;\leq\; \operatorname{rank}\begin{pmatrix} 0 & U \\ L & D \end{pmatrix}.$$

To see this, pick $\operatorname{rank}(U)$ columns of $\binom{U}{D}$ whose U-parts are linearly independent and $\operatorname{rank}(L)$ columns of $\binom{0}{L}$ whose L-parts are independent. If a linear combination of all these columns vanishes, its top part is a combination of the chosen U-parts only, so those coefficients vanish; then the bottom part is a combination of the chosen L-parts, so the remaining coefficients vanish too. Finally

$$\begin{pmatrix} 0 & U \\ L & D \end{pmatrix} = M - \begin{pmatrix} N & 0 \\ 0 & 0 \end{pmatrix} \quad\Longrightarrow\quad \operatorname{rank}\begin{pmatrix} 0 & U \\ L & D \end{pmatrix} \leq \operatorname{rank}(M) + \operatorname{rank}(N).$$

Adding up the three estimates gives the claim. □

Theorem 4.12 (Lovász [Lov90]). *Let f be Boolean with $M_f \neq 0$, and write $r = \operatorname{rank}(M_f)$, $k = C^1(f)$. Then*

$$\mathrm{D}(f) = O\big((1 + \log k)(1 + \log r)\big).$$

Consequently $\mathrm{D}(f) = O\big(\log \operatorname{rank}_+(M_f) \cdot \log \operatorname{rank}(M_f)\big)$ *(for* $\operatorname{rank}(M_f) \geq 2$*).*

Proof. Fix a 1-cover $R_1, \dots, R_k$, $R_i = A_i \times B_i$. The players maintain a common submatrix $M' = M_f|_{X' \times Y'}$ with $(x, y) \in X' \times Y'$; initially $X' = X$, $Y' = Y$. Restricted to $X' \times Y'$, the rectangles $R_i \cap (X' \times Y')$ still cover all 1-entries of M'. Let $r' = \operatorname{rank}(M')$.

If $r' \leq 3$, the players finish with the protocol of Proposition 4.9 applied to M' (at most 4 bits). Otherwise, for each i permute M' so that R_i becomes the top-left block:

$$M' = \begin{pmatrix} R_i & U_i \\ L_i & D_i \end{pmatrix},$$

where the top block-row consists of the rows $A_i \cap X'$ and the left block-column of the columns $B_i \cap Y'$. By Lemma 4.11,

$$\operatorname{rank}\begin{pmatrix} R_i & U_i \end{pmatrix} + \operatorname{rank}\begin{pmatrix} R_i \\ L_i \end{pmatrix} \leq r' + 3, \tag{4.1}$$

so at least one of the two is at most $(r' + 3)/2$.

1. Alice looks for an i with $x \in A_i$ and $\operatorname{rank}(R_i\ U_i) \le (r'+3)/2$. If she finds one, she sends i; both players restrict to the rows $A_i \cap X'$, i.e. $M' \leftarrow (R_i\ U_i)$.
2. Otherwise she sends a failure symbol, and Bob looks for a j with $y \in B_j$ and $\operatorname{rank}\binom{R_j}{L_j} \le (r'+3)/2$. If he finds one, he sends j and both restrict to the columns $B_j \cap Y'$.
3. If both fail, output 0.

Correctness. The input always stays inside the current submatrix. If both fail, then $f(x,y)=0$: otherwise some R_i contains (x,y), and by (4.1) either Alice could have chosen i (if $\operatorname{rank}(R_i\ U_i)$ is small) or Bob could have chosen it.

Cost. Each round in which someone succeeds replaces $r' \ge 4$ by at most $(r'+3)/2 \le 7r'/8$, so there are at most $\log_{8/7} r + 1$ rounds; each costs $O(1+\log k)$ bits. The final phase costs at most 4 bits. The last statement follows from Lemma 4.10. □

Remark 4.13. By Theorem 3.5, Theorem 4.12 can be read as $\mathrm{D}(f) = O(\mathrm{N}^1(f)\log\operatorname{rank}(M_f))$. Compared with the Aho–Ullman–Yannakakis theorem $\mathrm{D}(f) = O(\mathrm{N}^0(f)\mathrm{N}^1(f))$ (Theorem 3.8), one of the two nondeterministic complexities has been replaced by $\log\operatorname{rank}(M_f)$. Lovász [Lov90] states it in the complementary form, with $\mathrm{N}^0(f)$ and the rank of the complementary matrix, which is equivalent up to constants; see also the discussion in [LS09].

4.5 The log-rank conjecture

By Theorem 4.3 and Proposition 4.9, $\log\operatorname{rank}(M_f) \le \mathrm{D}(f) \le \operatorname{rank}(M_f)+1$. Lovász and Saks conjectured that the lower bound is close to the truth.

Conjecture 4.14 (Log-rank conjecture [LS88]). *There is a constant α such that every Boolean f satisfies*

$$\mathrm{D}(f) = O\big(\log^{\alpha}\operatorname{rank}(M_f)\big).$$

The conjecture is open. What is known:

- *Lower bound on α.* Göös, Pitassi and Watson [GPW15] constructed functions with $\mathrm{D}(f) \ge \widetilde{\Omega}(\log^2\operatorname{rank}(M_f))$ (via clique vs. independent set, whose rank is at most n by Example 4.7). Hence if the conjecture holds then $\alpha \ge 2$.
- *Upper bounds in terms of $r=\operatorname{rank}(M_f)$.* The trivial bound is $r+1$. Lovett [Lov16] proved $\mathrm{D}(f) = O(\sqrt{r}\log r)$, and Sudakov and Tomon [ST25] recently improved this to $O(\sqrt{r})$. Below we prove the slightly weaker bound $O(\sqrt{r}\log^2 r)$, which already contains the main idea (Theorem 4.16).

The proof combines the rank-splitting lemma with the following statement from extremal combinatorics, which we use without proof.

Lemma 4.15 (Large monochromatic rectangles [Lov16, GL14]). *There is an absolute constant $c>0$ such that every Boolean matrix M of rank $r \ge 1$ contains a monochromatic submatrix R with*

$$|R| \ \ge\ \delta(r)\,|M|, \qquad \text{where } \delta(r) := 2^{-c\sqrt{r}\log(r+1)}.$$

Theorem 4.16. *If f is Boolean with $r=\operatorname{rank}(M_f)\ge 1$, then $\mathrm{D}(f) = O(\sqrt{r}\log^2(r+1))$.*

Proof. Preprocessing. Identical rows of M_f can be merged: Alice replaces x by a fixed representative of its row class, which changes neither the answer nor the rank. Doing the same for columns, by Lemma 4.8 we may assume $|X|,|Y| \le 2^r$, so $|M_f| \le 4^r$.

A (deep) protocol tree. The players maintain a common submatrix $M' = M_f|_{X'\times Y'}$ containing (x,y), with $r' = \operatorname{rank}(M') \le r$. If M' is monochromatic, they output its colour. If $r' \le 3$ they finish with Proposition 4.9 (≤ 4 bits). Otherwise they take the monochromatic rectangle

$R = A \times B$ of M' given by Lemma 4.15 (both can compute it), with $|R| \geq \delta(r)|M'|$, and write $M' = \begin{pmatrix} R & U \\ L & D \end{pmatrix}$. By Lemma 4.11, the smaller of $\operatorname{rank}(R\ U)$ and $\operatorname{rank}\binom{R}{L}$ is at most $(r'+3)/2$.

- If $\operatorname{rank}(R\ U) \leq \operatorname{rank}\binom{R}{L}$, Alice sends the bit $[x \in A]$. On 1 they restrict to rows A (rank $\leq (r'+3)/2$); on 0 to rows $X' \setminus A$.
- Otherwise Bob sends $[y \in B]$, and they restrict to columns B or $Y' \setminus B$ accordingly.

Every round sends exactly one bit.

Counting. Fix a root-to-leaf path of the main phase.

- A bit 1 replaces $r' \geq 4$ by at most $(r'+3)/2 \leq 7r'/8$, so a path contains at most $k := \left\lceil \log_{8/7} r \right\rceil + 1 = O(\log r)$ ones.
- A bit 0 removes the rows A (or the columns B), hence at least $|R| \geq \delta(r)|M'|$ entries. After j zeros the current matrix has at most $4^r(1-\delta(r))^j \leq 4^r e^{-j\delta(r)}$ entries, and it is never empty, so a path contains at most $h_0 := r \ln 4/\delta(r) = 2^{O(\sqrt{r}\log(r+1))}$ zeros.

So the main phase has depth at most $h := k + h_0$. Map each leaf of the main phase to its root-to-leaf bit string, padded with zeros to length h. Distinct leaves have bit strings that are not prefixes of each other, so this map is injective, and every image has at most k ones. Therefore the main phase has at most

$$\sum_{i=0}^{k} \binom{h}{i} \leq (h+1)^k = 2^{O(\log r)\cdot O(\sqrt{r}\log(r+1))} = 2^{O(\sqrt{r}\log^2(r+1))}$$

leaves, and the final phase multiplies this by at most 2^4. Hence $C^P(f) \leq 2^{O(\sqrt{r}\log^2(r+1))}$, and the balancing theorem (Theorem 2.19) gives $\mathrm{D}(f) = O(\log C^P(f)) = O(\sqrt{r}\log^2(r+1))$. □

Remark 4.17. The protocol tree built in the proof is very deep; it is only its *number of leaves* that is small, and Theorem 2.19 converts this into a shallow protocol. Nisan and Wigderson [NW95] gave a more careful reduction showing that if every rank-r Boolean matrix contains a monochromatic rectangle of relative size $2^{-q(r)}$, then $\mathrm{D}(f) \leq O(\log^2 r) + \sum_{i\geq 0} O(q(r/2^i))$; with $q(r) = O(\sqrt{r}\log r)$ this removes one logarithmic factor and gives Lovett's $O(\sqrt{r}\log r)$ [Lov16]. Sudakov and Tomon [ST25] obtain $q(r) = O(\sqrt{r})$ and hence $\mathrm{D}(f) = O(\sqrt{r})$.

Looking ahead. This completes the picture for deterministic protocols: equality, greater-than, disjointness and inner product all need about n bits, and we have three families of lower-bound techniques (fooling sets and rectangle size, covers, rank) together with matching ways of turning rectangle structure back into protocols. In Part II the players are allowed to toss coins and to err with small probability. We shall see that this changes the landscape completely for some of these functions and not at all for others, and that explaining the difference requires new lower-bound techniques.

4.6 Notes and references

The rank lower bound is due to Mehlhorn and Schmidt [MS82]; see [KN97, Ch. 1] and [RY20, Ch. 2]. Theorem 4.12 is from Lovász's survey [Lov90]; it strengthens the bound $\mathrm{D}(f) = O(\mathrm{N}^0(f)\mathrm{N}^1(f))$ of Aho, Ullman and Yannakakis [AUY83] (Theorem 3.8). The log-rank conjecture was posed by Lovász and Saks [LS88]; Lovett's survey [Lov14] describes the state of the art around 2014 and its many equivalent formulations. The monochromatic-rectangle lemma (Lemma 4.15) is due to Lovett [Lov16] combined with Gavinsky and Lovett [GL14]; the improvement to $O(\sqrt{\operatorname{rank}})$ is due to Sudakov and Tomon [ST25]. Nonnegative rank and its connection to extended formulations of polytopes go back to Yannakakis [Yan91]; see [RY20, Ch. 12] and [FMP+15].

Part II

Randomized Communication

5 Randomized Protocols

Part I showed that equality, greater-than, disjointness and inner product all require about n bits of deterministic communication. In the deterministic model we think of an adversary who knows the protocol and picks the worst input. In the randomized model the adversary still knows the protocol and still picks the input, but cannot predict the outcomes of the players' coin flips; in exchange, the protocol is allowed to err with small probability. We will see that randomness can be extremely powerful (equality drops from $n+1$ bits to $O(1)$ or $O(\log n)$), and we will also see its limits.

5.1 Private and public coins

Definition 5.1 (Randomized protocols). A *private-coin protocol* is a protocol tree in which Alice's node functions are of the form $a_v(x, r_A)$ and Bob's of the form $b_v(y, r_B)$, where r_A and r_B are independent uniformly random strings (of some finite length) known only to Alice and to Bob, respectively. In a *public-coin protocol* both players see a common random string r and the node functions are $a_v(x, r)$ and $b_v(y, r)$. In both cases the cost is the maximum depth over all inputs and all random strings.

A randomized protocol computes f with *error* ε if for *every* input (x, y),

$$\Pr[\text{output} \neq f(x, y)] \leq \varepsilon,$$

the probability being over the random strings only. We write $\mathrm{R}_\varepsilon(f)$ and $\mathrm{R}^{\mathrm{pub}}_\varepsilon(f)$ for the minimum cost of a private-coin, resp. public-coin, protocol computing f with error ε.

For a fixed public string r, a public-coin protocol is a deterministic protocol tree T_r. So a public-coin protocol is the same thing as a probability distribution over deterministic protocol trees of bounded depth. Since public coins can simulate private ones (split r into two halves, one for each player),

$$\mathrm{R}^{\mathrm{pub}}_\varepsilon(f) \;\leq\; \mathrm{R}_\varepsilon(f) \;\leq\; \mathrm{D}(f).$$

A protocol has *one-sided error* if it never errs on inputs with $f(x, y) = 1$ (or never on inputs with $f(x, y) = 0$). Often one only asks the protocol to be correct with probability $1/2 + \Omega(1)$, that is, with error bounded away from $1/2$. Every fixed constant $\varepsilon < 1/2$ leads to the same complexity up to a constant factor, because errors can be reduced by repetition (Lemma 5.2 below); we set

$$\mathrm{R}(f) := \mathrm{R}_{1/3}(f), \qquad \mathrm{R}^{\mathrm{pub}}(f) := \mathrm{R}^{\mathrm{pub}}_{1/3}(f).$$

Lemma 5.2 (Error reduction). *Let f be Boolean, $0 \leq \varepsilon < 1/2$ and $0 < \varepsilon' < 1/2$. Running a protocol with error ε independently k times and outputting the majority answer gives error at most $e^{-2k(1/2-\varepsilon)^2}$. Hence*

$$\mathrm{R}_{\varepsilon'}(f) \leq \left\lceil \frac{\ln(1/\varepsilon')}{2(1/2-\varepsilon)^2} \right\rceil \cdot \mathrm{R}_\varepsilon(f),$$

and the same holds for public coins. For a protocol with one-sided error ε, k repetitions (outputting 1 only if all runs say 1, in the case where errors occur only on 0-inputs) give one-sided error ε^k.

Proof. Let C be the number of runs that answer correctly. It is a sum of k independent indicators, each with mean at least $1-\varepsilon$. The majority is wrong only if $C \leq k/2$, i.e. $C - \mathbb{E}\,C \leq -k(1/2-\varepsilon)$; by Hoeffding's inequality (Lemma A.1) this has probability at most $e^{-2k(1/2-\varepsilon)^2}$. The one-sided statement is clear. □

5.2 Equality

Recall $\mathrm{D}(\mathrm{EQ}_n) = n+1$ (Example 2.10). Randomness changes this dramatically.

5.2.1 Fingerprints modulo a random prime

Fact 5.3. *The k-th prime number p_k satisfies $p_k = O(k \log k)$. (This follows from the prime number theorem, or from Chebyshev's elementary estimates.)*

Protocol 5.4 (Equality via primes). Let $p_1 < p_2 < \cdots < p_{2n}$ be the first $2n$ primes. Alice views x as an integer in $[0, 2^n)$, picks $i \in [2n]$ uniformly with her private coins, and sends i and $x \bmod p_i$. Bob outputs 1 iff $x \equiv y \pmod{p_i}$, and sends the answer to Alice.

Proposition 5.5. *Protocol 5.4 costs $O(\log n)$ bits and has one-sided error less than $1/2$. Consequently $\mathrm{R}(\mathrm{EQ}_n) = O(\log n)$.*

Proof. If $x = y$ the protocol is always correct. If $x \neq y$, it errs iff $p_i \mid x - y$. Since $0 < |x-y| < 2^n$ and a product of j distinct primes is at least 2^j, the integer $x - y$ has fewer than n distinct prime divisors. So at most $n-1$ of the $2n$ candidate primes divide it, and the error probability is at most $(n-1)/(2n) < 1/2$. By Fact 5.3 the index and the residue take $O(\log n)$ bits. Two independent repetitions bring the error below $1/4$ (Lemma 5.2). □

5.2.2 Public coins: random inner products

Proposition 5.6. $\mathrm{R}^{\mathrm{pub}}_{\varepsilon}(\mathrm{EQ}_n) \leq \lceil \log(1/\varepsilon) \rceil + 1$ *for every $\varepsilon > 0$. In particular* $\mathrm{R}^{\mathrm{pub}}(\mathrm{EQ}_n) = O(1)$.

Proof. Using the public string, pick $k = \lceil \log(1/\varepsilon) \rceil$ independent uniform vectors $r^{(1)}, \ldots, r^{(k)} \in \{0,1\}^n$. Alice sends the k bits $\langle x, r^{(j)} \rangle \bmod 2$; Bob compares them with $\langle y, r^{(j)} \rangle \bmod 2$ and announces whether all agree. If $x = y$ the answer is always correct. If $x \neq y$, let $z = x \oplus y \neq 0$ and pick i with $z_i = 1$. For each j, flipping the i-th coordinate of $r^{(j)}$ is a bijection that changes $\langle z, r^{(j)} \rangle$, so $\Pr[\langle z, r^{(j)} \rangle = 0] = 1/2$ exactly. The k tests all pass with probability $2^{-k} \leq \varepsilon$. □

A single test ($k = 1$) costs two bits and has one-sided error exactly $1/2$. This one-sided test will be used repeatedly below.

5.2.3 Error-correcting codes

The prime-number protocol is an instance of a general principle: encode x by a code with large distance, and compare a random coordinate.

Definition 5.7. A map $C \colon \{0,1\}^n \to \Sigma^m$ with $|\Sigma| = 2^k$ is a code with *agreement* at most δ if for all $x \neq x'$, $|\{i \in [m] : C(x)_i = C(x')_i\}| \leq \delta m$.

Protocol 5.8 (Equality via a code). Fix such a code C, known to both players. Alice picks $i \in [m]$ uniformly at random and sends i and $C(x)_i$; Bob outputs 1 iff $C(x)_i = C(y)_i$ and sends the answer back. The cost is $\lceil \log m \rceil + k + 1$ and the error is one-sided and at most δ.

Proposition 5.9. *For every $k \geq 6$ there is a code $C \colon \{0,1\}^n \to \Sigma^m$ with $|\Sigma| = 2^k$, agreement at most $\delta = 2^{-k/2}$, and $m = \left\lfloor 8n 2^{k/2}/k \right\rfloor + 1$. Consequently $\mathrm{R}_{2^{-k/2}}(\mathrm{EQ}_n) = O(\log n + k)$.*

Proof. Choose all $2^n \cdot m$ symbols $C(x)_i$ independently and uniformly. Fix $x \neq x'$; each coordinate agrees independently with probability 2^{-k}. Let $s = \lfloor \delta m \rfloor + 1$. By a union bound over the sets of s coordinates,

$$\Pr[\text{at least } s \text{ agreements}] \leq \binom{m}{s} 2^{-ks} \leq \left(\frac{em}{s}\right)^s 2^{-ks} \leq \left(\frac{e}{\delta} 2^{-k}\right)^s = \left(e\, 2^{-k/2}\right)^s \leq 2^{-ks/4},$$

using $s \geq \delta m$ and $e \leq 2^{k/4}$ for $k \geq 6$. As $ks/4 \geq k\delta m/4 > 2n$ by the choice of m, this is less than 2^{-2n}, and a union bound over the fewer than 2^{2n} pairs shows that some code works. Plugging it into Protocol 5.8 gives cost $O(\log m + k) = O(\log n + k)$. □

Remark 5.10. Explicit codes also work. For example (Reed–Solomon), split x into $d \approx n/\log q$ blocks, regard them as the coefficients of a polynomial P_x of degree less than d over a prime field $\mathbb{F}_q$ with $q \approx n$, and let $C(x) = (P_x(\alpha))_{\alpha \in \mathbb{F}_q}$. Distinct polynomials of degree $< d$ agree on fewer than d points, so the agreement is at most $d/q = O(1/\log n)$ and the protocol costs $O(\log n)$.

We will show in Section 5.5 that $\mathrm{R}(\mathrm{EQ}_n) = \Omega(\log n)$; so for equality the private-coin and the public-coin complexities really are different.

5.3 Greater-than

For $x, y \in \{0,1\}^n$ let $\ell(x,y) \in \{0,1,\dots,n\}$ be the length of their longest common prefix. If $\ell = \ell(x,y) < n$ then x and y first differ at position $\ell + 1$ and

$$\mathrm{GT}(x,y) = x_{\ell+1};$$

if $\ell = n$ then $x = y$ and $\mathrm{GT}(x,y) = 0$. So computing GT reduces to finding ℓ, and for each t the predicate $\ell \geq t$ is an *equality test* on prefixes:

$$\ell(x,y) \geq t \iff x_1 \cdots x_t = y_1 \cdots y_t.$$

Binary search. With exact equality tests, binary search finds ℓ after $\lceil \log(n+1) \rceil$ tests. With randomized tests each test may fail, and to make all of them succeed simultaneously (union bound) each must have error $O(1/\log n)$:

- with public coins, each test costs $O(\log \log n)$ bits (Proposition 5.6), giving $\mathrm{R}^{\mathrm{pub}}(\mathrm{GT}_n) = O(\log n \log \log n)$;
- with private coins, each test costs $O(\log n)$ bits (Proposition 5.9 with $k = O(\log \log n)$), giving $\mathrm{R}(\mathrm{GT}_n) = O(\log^2 n)$.

Noisy binary search. The $\log \log n$ overhead comes from insisting that *every* test be correct. A better idea is to let the search make mistakes and *detect and undo* them, which turns binary search into a random walk with a drift towards the right answer.

Theorem 5.11. $\mathrm{R}^{\mathrm{pub}}_{\varepsilon}(\mathrm{GT}_n) = O(\log n + \log(1/\varepsilon))$. *In particular* $\mathrm{R}^{\mathrm{pub}}(\mathrm{GT}_n) = O(\log n)$.

Proof. The search tree. Let $\mathcal{T}$ be the balanced binary search tree on $\{0,1,\dots,n\}$: the root is the interval $[0,n]$, an interval $[a,b]$ with $a < b$ has children $[a,m]$ and $[m+1,b]$ where $m = \lfloor (a+b)/2 \rfloor$, and the leaves are the singletons $[s,s]$. Its depth is $d \leq \lceil \log(n+1) \rceil$. Fix a length T (chosen below) and hang below every leaf $[s,s]$ a path of T further nodes, all labelled s; call it the *chain* of s.

Tests. A *test* is the one-sided public-coin equality test of Proposition 5.6 applied to two prefixes, repeated 5 times; it costs 6 bits and errs with probability at most $2^{-5} < 1/24$. Using such tests, the players can check "$\ell \in [a,b]$" (test $\ell \geq a$, and if $b < n$ test that $\ell \geq b+1$ fails).

The walk. Start at the root and perform T steps. At the current node v:

- if $v = [a,b]$ is an internal node of $\mathcal{T}$: check $\ell \in [a,b]$; if the check fails, move to the parent (stay if v is the root); otherwise test $\ell \geq m+1$ and move to the right child $[m+1,b]$ if it passes and to the left child $[a,m]$ if it fails;
- if v is a leaf $[s,s]$ or a chain node labelled s: check $\ell \in [s,s]$; if it passes move one step down the chain (stay at the bottom), otherwise move one step up.

After T steps, if the walk is at a node labelled s (a leaf or a chain node), the players output 0 if $s = n$ and otherwise Alice sends x_{s+1}, which is the output; at an internal node they output 0.

Analysis. Let $\ell^\star = \ell(x, y)$ and let τ be the bottom node of the chain of $\ell^\star$. For every node $v \neq \tau$ call the first edge of the shortest path from v to τ the *correct move*. One checks directly that, if all tests in a step give the right answers, the walk makes the correct move: from an internal node whose interval does not contain $\ell^\star$ it goes up; from one containing $\ell^\star$ it goes to the child containing $\ell^\star$; from a wrong chain it goes up; on the correct chain it goes down. A step uses at most three tests, so, whatever happened before, it makes the correct move with probability at least $1 - 3/24 = 7/8$.

Let C be the number of steps that make the correct move. The starting distance to τ is $d^\star + T$, where $d^\star \in [1, d]$ is the depth of the leaf $[\ell^\star, \ell^\star]$; since this exceeds T, the walk never reaches τ. Each correct move decreases the distance to τ by one, and any other move increases it by at most one. Hence after T steps the distance is at most $d^\star + T - C + (T - C)$. The nodes at distance at most T from τ are exactly the leaf $[\ell^\star, \ell^\star]$ and its chain, so the output is correct whenever $d^\star + 2T - 2C \leq T$, i.e. whenever $C \geq (T + d^\star)/2$.

Choose $T = \max(4d, \lceil 8 \ln(1/\varepsilon) \rceil)$. Then $(T + d^\star)/2 \leq 5T/8$, and since each step is correct with probability at least $7/8$ given the past, Lemma A.2 gives

$$\Pr[C < 5T/8] \leq e^{-2(1/4)^2 T} = e^{-T/8} \leq \varepsilon.$$

The cost is at most $18T + 1 = O(\log n + \log(1/\varepsilon))$. □

Combined with Newman's theorem below, Theorem 5.11 gives $\mathrm{R}(\mathrm{GT}_n) = O(\log n)$.

5.4 Newman's theorem: public versus private coins

A public-coin protocol may use a very long random string. Newman showed that a short one suffices, and that a short random string can be sent by one player to the other.

Theorem 5.12 (Newman [New91]). *Let $f\colon \{0,1\}^n \times \{0,1\}^n \to Z$, $\varepsilon \geq 0$ and $\delta > 0$. Then*

$$\mathrm{R}_{\varepsilon+\delta}(f) \;\leq\; \mathrm{R}^{\mathrm{pub}}_{\varepsilon}(f) + \log n + 2\log(1/\delta) + O(1).$$

Proof. Fix a public-coin protocol P of cost $c = \mathrm{R}^{\mathrm{pub}}_{\varepsilon}(f)$ and error ε. For an input (x, y) and a random string r let $Z(x, y, r) = 1$ if P errs on (x, y) with randomness r; so $\mathbb{E}_r\, Z(x, y, r) \leq \varepsilon$ for every (x, y).

Let $t = \lceil n/\delta^2 \rceil$ and draw $r_1, \ldots, r_t$ independently. For a fixed (x, y), Hoeffding's inequality gives

$$\Pr\Big[\frac{1}{t}\sum_{j=1}^{t} Z(x, y, r_j) > \varepsilon + \delta\Big] \leq e^{-2\delta^2 t} \leq e^{-2n}.$$

By the union bound over the 4^n inputs, the probability that some input is bad is at most $(4/e^2)^n < 1$. So there exist *fixed* strings $r_1, \ldots, r_t$ such that for every input, at most an $(\varepsilon + \delta)$ fraction of them make P err.

The new private-coin protocol hard-wires $r_1, \ldots, r_t$. Alice picks $j \in [t]$ uniformly with her private coins, sends j using $\lceil \log t \rceil \leq \log n + 2\log(1/\delta) + 2$ bits, and then both run P with randomness r_j. On every input the error is at most $\varepsilon + \delta$. □

Corollary 5.13. *For Boolean f on n-bit inputs, $\mathrm{R}(f) \leq O(\mathrm{R}^{\mathrm{pub}}(f) + \log n)$. In particular $\mathrm{R}(\mathrm{EQ}_n) = O(\log n)$ and $\mathrm{R}(\mathrm{GT}_n) = O(\log n)$.*

Proof. Apply Theorem 5.12 with $\varepsilon = 1/4$ and $\delta = 1/12$ (after reducing the error of a public-coin protocol to $1/4$ by Lemma 5.2), and use Proposition 5.6 and Theorem 5.11. □

5.5 Private coins save at most exponentially

Theorem 5.14. *Let f be Boolean, $0 \le \varepsilon < 1/2$, and $c = \mathrm{R}_\varepsilon(f)$. Then*

$$\mathrm{D}(f) \;\le\; 2^c\Big(c + \Big\lceil \log \tfrac{1}{1/2-\varepsilon} \Big\rceil + 2\Big) + 1.$$

Consequently $\mathrm{R}_\varepsilon(f) \ge \log \mathrm{D}(f) - \log\log \mathrm{D}(f) - O_\varepsilon(1)$, i.e. $\mathrm{R}(f) = \Omega(\log \mathrm{D}(f))$.

Proof. Fix a private-coin protocol of cost c and error ε; its tree has at most 2^c leaves. For a leaf ℓ, let $p_\ell(x)$ be the probability over r_A that at *every Alice node* on the root-to-ℓ path, $a_v(x, r_A)$ points along the path; define $q_\ell(y)$ similarly for Bob's nodes and r_B. The computation reaches ℓ iff both events happen, and r_A, r_B are independent, so

$$\Pr[\text{the protocol reaches } \ell \text{ on } (x,y)] = p_\ell(x)\, q_\ell(y). \tag{5.1}$$

Let L_1 be the set of leaves labelled 1 and $S(x,y) = \sum_{\ell \in L_1} p_\ell(x) q_\ell(y)$ the acceptance probability. Then $S \ge 1-\varepsilon$ if $f(x,y) = 1$ and $S \le \varepsilon$ if $f(x,y) = 0$.

Deterministic protocol: for every leaf ℓ, Alice sends $p_\ell(x)$ rounded down to a multiple of 2^{-k}, where $k = c + \left\lceil \log \frac{1}{1/2-\varepsilon} \right\rceil + 1$; this takes $k+1$ bits per leaf. Bob computes the corresponding approximation $\tilde{S}$ of S. Since every $q_\ell(y) \le 1$,

$$|\tilde{S} - S| \le \sum_{\ell \in L_1} 2^{-k} q_\ell(y) \le 2^c 2^{-k} \le \tfrac{1}{2}\big(\tfrac{1}{2} - \varepsilon\big).$$

So $\tilde{S} > 1/2$ if $f(x,y) = 1$ and $\tilde{S} < 1/2$ if $f(x,y) = 0$. Bob announces the answer with one more bit. □

Corollary 5.15. *$\mathrm{R}(\mathrm{EQ}_n)$, $\mathrm{R}(\mathrm{GT}_n)$, $\mathrm{R}(\mathrm{IP}_n)$ and $\mathrm{R}(\mathrm{DISJ}_n)$ are all $\Omega(\log n)$. Hence $\mathrm{R}(\mathrm{EQ}_n) = \Theta(\log n)$ and $\mathrm{R}(\mathrm{GT}_n) = \Theta(\log n)$.*

Remark 5.16. The factorisation (5.1) is the only place where private coins were used, and the theorem is false for public coins: $\mathrm{R}^{\mathrm{pub}}(\mathrm{EQ}_n) = O(1)$ while $\log \mathrm{D}(\mathrm{EQ}_n) \approx \log n$. The same product structure of private-coin transcripts reappears, as the "cut-and-paste" property, in the disjointness lower bound (Lemma 10.13).

5.6 Summary

	EQ_n	GT_n	IP_n	DISJ_n
D	$n+1$	$\Theta(n)$	$\Theta(n)$	$n+1$
R	$\Theta(\log n)$	$\Theta(\log n)$	$\Omega(\log n)$	$\Omega(\log n)$
$\mathrm{R}^{\mathrm{pub}}$	$\Theta(1)$	$O(\log n)$	?	?

The question marks will be resolved in Chapter 7 (IP) and Chapter 10 (DISJ): both need $\Omega(n)$ bits even with public coins. In Chapter 7 we also show $\mathrm{R}^{\mathrm{pub}}(\mathrm{GT}_n) = \Omega(\log n)$. All of these are lower bounds against *public-coin* protocols, for which Theorem 5.14 says nothing. The first step towards them, taken in the next chapter, is to replace the randomness of the protocol by randomness of the input.

5.7 Notes and references

Randomized communication complexity is treated in [KN97, Ch. 3] and [RY20, Ch. 3]. The prime-fingerprint protocol for equality is folklore; the public-coin protocol is the standard "random parity" hash. Theorem 5.12 is from Newman [New91]. Theorem 5.14 appears in [KN97, Ch. 3]. The $O(\log n)$ protocol for greater-than is an instance of noisy binary search, studied in general by Feige, Raghavan, Peleg and Upfal [FRPU94]; the matching $\Omega(\log n)$ lower bound for public coins is proved in Section 7.3 (see also Viola [Vio15], who determines the complexity of greater-than and of related problems more precisely).

6 Distributional Complexity and Yao's Principle

Lower bounds for randomized protocols are hard to prove directly: a randomized protocol is a probability distribution over deterministic protocols, and we must argue about all such distributions at once. Yao's principle replaces the randomness in the protocol by randomness in the *input*: to show that randomized protocols need many bits, it suffices to exhibit one input distribution on which every cheap *deterministic* protocol errs often. Deterministic protocols partition the input space into rectangles, so all the rectangle machinery of Part I becomes available again.

6.1 Distributional complexity

Definition 6.1 (Distributional complexity). Let μ be a probability distribution on $X \times Y$ and $0 \le \varepsilon < 1/2$. The *distributional complexity* $\mathrm{D}^{\mu}_{\varepsilon}(f)$ is the minimum cost of a *deterministic* protocol T with

$$\Pr_{(x,y)\sim\mu}\left[T(x,y) \neq f(x,y)\right] \le \varepsilon.$$

The protocol is now allowed to be wrong on some inputs, as long as these inputs have small total weight under μ. Whether $\mathrm{D}^{\mu}_{\varepsilon}(f)$ is large depends heavily on μ.

Example 6.2 *(Greater-than under the uniform distribution).* Let μ be uniform on $\{0,1\}^n \times \{0,1\}^n$. Alice sends her most significant bit x_1, and Bob outputs 1 iff $x_1 = 1$ and $y_1 = 0$ (and sends this bit to Alice). If $x_1 \neq y_1$ the output is correct. If $x_1 = y_1$, the protocol outputs 0, which is correct iff $x \le y$, an event of probability at least $1/2$ given $x_1 = y_1$. So the success probability is at least $\frac{1}{2}\cdot 1 + \frac{1}{2}\cdot\frac{1}{2} = \frac{3}{4}$ and

$$\mathrm{D}^{\mu}_{1/4}(\mathrm{GT}_n) \le 2.$$

Sending k bits instead of one reduces the error to below 2^{-k-1}. See Figure 6.1.

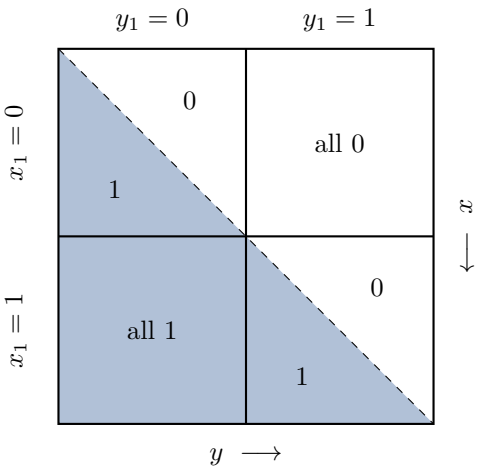


Figure 6.1: The matrix of GT_n (rows x increase downwards, columns y to the right; shaded entries are 1). The protocol of Example 6.2 outputs 1 exactly on the bottom-left quadrant. It is correct on the two off-diagonal quadrants and on about half of each diagonal quadrant.

So the uniform distribution is easy for greater-than; a lower bound will need a cleverly chosen distribution (Section 7.3). The following direction is the one used for lower bounds and holds for every μ.

Proposition 6.3. *For every distribution μ, $\mathrm{D}^{\mu}_{\varepsilon}(f) \le \mathrm{R}^{\mathrm{pub}}_{\varepsilon}(f)$.*

Proof. Let P be a public-coin protocol of cost c and error ε; for each fixed string r it is a deterministic protocol T_r of cost at most c. For every input, $\Pr_r[T_r(x,y) \neq f(x,y)] \le \varepsilon$. Averaging over $(x,y) \sim \mu$ and exchanging the order of expectation,

$$\mathop{\mathbb{E}}_{r}\Big[\Pr_{(x,y)\sim\mu}[T_r(x,y) \neq f(x,y)]\Big] \le \varepsilon,$$

so some fixed r has $\Pr_{\mu}[T_r(x,y) \neq f(x,y)] \le \varepsilon$. □

6.2 The minimax theorem

Write $\Delta_k = \{p \in \mathbb{R}^k_{\ge 0} : \sum_i p_i = 1\}$ for the set of probability distributions on $[k]$.

Theorem 6.4 (von Neumann [vN28]). *For every real $m \times n$ matrix A,*

$$\min_{p\in\Delta_m}\max_{q\in\Delta_n} p^{\top}Aq \;=\; \max_{q\in\Delta_n}\min_{p\in\Delta_m} p^{\top}Aq.$$

Interpretation: in a zero-sum game where the row player pays $A[i,j]$ to the column player, $p^{\top}Aq$ is the expected payment when the players use mixed strategies p and q. The inequality "$\ge$" is easy (the player who commits second has an advantage); the content of the theorem is that committing first to a *mixed* strategy costs nothing. It follows from linear programming duality. Note also that for fixed p the inner maximum over q is attained at a point mass, since $p^{\top}Aq$ is linear in q.

6.3 Yao's minimax principle

Theorem 6.5 (Yao's principle [Yao77]). *For every f and every $0 \le \varepsilon < 1/2$,*

$$\mathrm{R}^{\mathrm{pub}}_{\varepsilon}(f) = \max_{\mu} \mathrm{D}^{\mu}_{\varepsilon}(f).$$

Proof. "$\ge$" is Proposition 6.3. For "$\le$", let $c = \max_{\mu} \mathrm{D}^{\mu}_{\varepsilon}(f)$. Consider the zero-sum game in which the row player chooses a deterministic protocol tree T of depth at most c (there are finitely many, since X, Y, Z are finite), the column player chooses an input (x,y), and the row player pays

$$A[T,(x,y)] = \mathbf{1}[T(x,y) \neq f(x,y)].$$

A mixed strategy of the column player is a distribution μ on inputs; a mixed strategy of the row player is a distribution ν on protocol trees of depth at most c, i.e. a public-coin protocol of cost at most c.

By the definition of c, for every μ there is a deterministic T of depth at most c with error at most ε under μ. Hence

$$\max_{\mu}\min_{\nu}\mathop{\mathbb{E}}_{T\sim\nu,(x,y)\sim\mu} A[T,(x,y)] \le \max_{\mu}\min_{T}\Pr_{\mu}[T(x,y) \neq f(x,y)] \le \varepsilon.$$

By Theorem 6.4 there is a $\nu^{\star}$ with $\max_{\mu}\mathbb{E}_{T\sim\nu^{\star},(x,y)\sim\mu} A[T,(x,y)] \le \varepsilon$. Taking μ to be the point mass on any fixed input (x,y) shows $\Pr_{T\sim\nu^{\star}}[T(x,y) \neq f(x,y)] \le \varepsilon$ for every input. So $\nu^{\star}$ is a public-coin protocol of cost at most c and error at most ε. □

Remark 6.6. Two comments.

1. For lower bounds only the easy direction is needed: *to show* $\mathrm{R}^{\mathrm{pub}}_{\varepsilon}(f) \ge c$, *exhibit a distribution* μ *with* $\mathrm{D}^{\mu}_{\varepsilon}(f) \ge c$. Choosing μ well is the art; the uniform distribution is often, but not always, a good choice (Example 6.2).

2. The proof never used the structure of protocol trees, only that for each cost bound there are finitely many deterministic protocols. So the same principle holds for restricted classes of protocols, for example *one-way* protocols, in which only Alice speaks. We use this in Section 10.2.

6.4 Notes and references

The minimax theorem is due to von Neumann [vN28]. Yao [Yao77] introduced the minimax principle for randomized algorithms; its communication version is in [KN97, Ch. 3]. In [RY20, Ch. 3] the same idea appears as the statement that a randomized protocol with small error yields a large *nearly* monochromatic rectangle under any distribution, which is the starting point for the discrepancy and corruption methods.

7 The Discrepancy Method

A deterministic protocol that is always correct partitions the communication matrix into *monochromatic* rectangles. A deterministic protocol that is only correct with probability $1-\varepsilon$ under μ still partitions the matrix into rectangles, but they need not be monochromatic. If ε is noticeably smaller than $1/2$, however, the leaf rectangles must be *biased* on average towards their output value. If every rectangle is nearly balanced between 0- and 1-inputs, there must be many leaves. This is the discrepancy method.

7.1 Discrepancy and the lower bound

Definition 7.1 (Discrepancy). Let $f\colon X \times Y \to \{0,1\}$ and let μ be a distribution on $X \times Y$. The discrepancy of a rectangle R is

$$\operatorname{disc}_\mu(f,R) = \Bigl|\sum_{(x,y)\in R} \mu(x,y)(-1)^{f(x,y)}\Bigr| = \bigl|\mu(R\cap f^{-1}(0)) - \mu(R\cap f^{-1}(1))\bigr|,$$

and the discrepancy of f is $\operatorname{disc}_\mu(f) = \max_R \operatorname{disc}_\mu(f,R)$, the maximum over all rectangles.

Two remarks help with intuition. If R is monochromatic then $\operatorname{disc}_\mu(f,R) = \mu(R)$, so small discrepancy implies in particular that all monochromatic rectangles are small (compare Lemma 2.13). Writing $\mathbf{1}_R$ for the indicator of R,

$$\operatorname{disc}_\mu(f,R) = \Bigl|\mathop{\mathbb{E}}_{(x,y)\sim\mu}\bigl[\mathbf{1}_R(x,y)(-1)^{f(x,y)}\bigr]\Bigr|,$$

which is the form we will use in computations.

Theorem 7.2 (Discrepancy lower bound). *For every Boolean f, distribution μ and $0 \le \varepsilon < 1/2$,*

$$\mathrm{D}^\mu_\varepsilon(f) \;\ge\; \log \frac{1-2\varepsilon}{\operatorname{disc}_\mu(f)}.$$

Consequently, by Theorem 6.5, $\mathrm{R}^{\mathrm{pub}}_\varepsilon(f) \ge \log \frac{1-2\varepsilon}{\operatorname{disc}_\mu(f)}$ for every μ.

Proof. Let T be a deterministic protocol of cost c with error at most ε under μ. Its leaves give a partition of $X \times Y$ into rectangles $R_1,\dots,R_t$, $t \le 2^c$, with outputs $z_1,\dots,z_t$. Then

$$\begin{aligned} 1-2\varepsilon \le \Pr_\mu[T=f] - \Pr_\mu[T\ne f] &= \sum_{i=1}^{t}\Bigl(\mu\bigl(R_i\cap f^{-1}(z_i)\bigr) - \mu\bigl(R_i\cap f^{-1}(1-z_i)\bigr)\Bigr) \\ &\le \sum_{i=1}^{t} \operatorname{disc}_\mu(f,R_i) \le 2^c \operatorname{disc}_\mu(f). \end{aligned}$$

Rearranging gives $c \ge \log((1-2\varepsilon)/\operatorname{disc}_\mu(f))$. □

So to prove a lower bound of $\Omega(k)$ it suffices to find one distribution under which every rectangle has discrepancy $2^{-\Omega(k)}$.

7.2 Inner product

We use the following facts from linear algebra: for a real matrix H, $\|H\| = \max_{\|v\|=1} \|Hv\|$ is the largest singular value of H, and $|u^\top H v| \le \|u\|\,\|H\|\,\|v\|$ by the Cauchy–Schwarz inequality.

Lemma 7.3 (Lindsey's lemma). *Let $H[x,y] = (-1)^{\mathrm{IP}(x,y)}$ for $x, y \in \{0,1\}^n$. Then $HH^\top = 2^n I$, so $\|H\| = 2^{n/2}$, and for all $A, B \subseteq \{0,1\}^n$,*

$$\Big|\sum_{x\in A}\sum_{y\in B}(-1)^{\mathrm{IP}(x,y)}\Big| \le \sqrt{|A|\,|B|\,2^n}.$$

Proof. $(HH^\top)[x,x'] = \sum_z (-1)^{\langle x,z\rangle + \langle x',z\rangle} = \sum_z (-1)^{\langle x \oplus x', z\rangle}$. For $x = x'$ every term is 1. For $x \neq x'$ pick i with $x_i \neq x'_i$; pairing each z with the string z' obtained by flipping z_i changes the sign of the term, so the sum is 0. Thus $HH^\top = 2^n I$, all singular values of H equal $2^{n/2}$, and

$$|\mathbf{1}_A^\top H \mathbf{1}_B| \le \|\mathbf{1}_A\|\,\|H\|\,\|\mathbf{1}_B\| = \sqrt{|A|}\cdot 2^{n/2}\cdot\sqrt{|B|}. \qquad \square$$

Theorem 7.4 ([CG88]). *Under the uniform distribution U, $\mathrm{disc}_U(\mathrm{IP}_n) \le 2^{-n/2}$. Hence*

$$\mathrm{R}^{\mathrm{pub}}_\varepsilon(\mathrm{IP}_n) \ge \frac{n}{2} - \log\frac{1}{1-2\varepsilon}, \qquad \text{so} \qquad \mathrm{R}(\mathrm{IP}_n) = \Theta(n), \quad \mathrm{R}^{\mathrm{pub}}(\mathrm{IP}_n) = \Theta(n).$$

Proof. For a rectangle $A \times B$, Lemma 7.3 gives

$$\mathrm{disc}_U(\mathrm{IP}, A\times B) = 4^{-n}|\mathbf{1}_A^\top H \mathbf{1}_B| \le 4^{-n}\sqrt{2^n\cdot 2^n\cdot 2^n} = 2^{-n/2}.$$

Apply Theorem 7.2; the upper bounds are trivial. $\square$

Note how much stronger this is than the rank bound: IP_n needs $\Omega(n)$ bits even with public coins and constant error, whereas EQ_n, whose matrix has essentially the same rank, needs only $O(1)$.

7.3 Greater-than

As Example 6.2 shows, the uniform distribution is useless for GT_n. The hard distribution must concentrate on pairs (x, y) that share a long common prefix, with the length of the prefix spread out.

Definition 7.5 (A hard distribution for GT). For $x \in \{0,1\}^n$ and $i \in [n]$ let $x^{(i)}$ be the string obtained from x by flipping the bits $i, i+1, \dots, n$. Let μ be the distribution of $(X, X^{(I)})$ where X is uniform on $\{0,1\}^n$ and I is uniform on $[n]$, independent of X.

The strings x and $x^{(i)}$ first differ at position i, where x has x_i and $x^{(i)}$ has $1 - x_i$. Hence

$$\mathrm{GT}(x, x^{(i)}) = x_i, \qquad (-1)^{\mathrm{GT}(x,x^{(i)})} = (-1)^{x_i}.$$

Theorem 7.6. $\mathrm{disc}_\mu(\mathrm{GT}_n) \le 1/\sqrt{n}$. *Consequently* $\mathrm{R}^{\mathrm{pub}}_\varepsilon(\mathrm{GT}_n) \ge \frac{1}{2}\log n - \log\frac{1}{1-2\varepsilon}$, *and*

$$\mathrm{R}^{\mathrm{pub}}(\mathrm{GT}_n) = \Theta(\log n), \qquad \mathrm{R}(\mathrm{GT}_n) = \Theta(\log n).$$

Proof. Fix a rectangle $A \times B$ and set $g(x) = \mathbb{E}_I\big[\mathbf{1}_B(x^{(I)})(-1)^{x_I}\big]$. Then

$$\mathrm{disc}_\mu(\mathrm{GT}, A\times B) = \big|\mathop{\mathbb{E}}_X[\mathbf{1}_A(X)\,g(X)]\big| \le \sqrt{\mathop{\mathbb{E}}_X[\mathbf{1}_A(X)^2]}\sqrt{\mathop{\mathbb{E}}_X[g(X)^2]} \le \sqrt{\mathop{\mathbb{E}}_X[g(X)^2]}$$

by Cauchy–Schwarz. Expanding the square with two independent uniform indices $I, J \in [n]$,

$$\mathop{\mathbb{E}}_{X}[g(X)^2] = \mathop{\mathbb{E}}_{I,J}\Big[\underbrace{\mathop{\mathbb{E}}_{X}[\mathbf{1}_B(X^{(I)})\mathbf{1}_B(X^{(J)})(-1)^{X_I+X_J}]}_{=:\gamma(I,J)}\Big].$$

If $I = J$, then $|\gamma(I,J)| \le 1$. Suppose $I = i < j = J$ (the case $i > j$ is symmetric). Let σ be the map that flips the bits $i, \dots, j-1$ of a string; it is a bijection of $\{0,1\}^n$ (an involution). Flipping bits $i, \dots, n$ of $\sigma(x)$ flips bits $i, \dots, j-1$ twice and bits $j, \dots, n$ once, so $\sigma(x)^{(i)} = x^{(j)}$; similarly $\sigma(x)^{(j)} = x^{(i)}$. Hence the product $\mathbf{1}_B(x^{(i)})\mathbf{1}_B(x^{(j)})$ is the same for x and $\sigma(x)$. On the other hand σ flips x_i but not x_j, so $(-1)^{x_i+x_j}$ changes sign. Pairing x with $\sigma(x)$ shows $\gamma(i,j) = 0$. Therefore

$$\mathop{\mathbb{E}}_{X}[g(X)^2] \le \Pr[I = J] = \frac{1}{n},$$

and $\mathrm{disc}_\mu(\mathrm{GT}, A \times B) \le 1/\sqrt{n}$. The lower bounds follow from Theorem 7.2, and the upper bounds from Theorem 5.11 and Corollary 5.13. □

7.4 Limitations, and a summary

Discrepancy cannot prove strong lower bounds for functions that have a small cover by monochromatic rectangles of *one* colour. Disjointness is the prime example.

Proposition 7.7. *For every distribution μ, $\mathrm{disc}_\mu(\mathrm{DISJ}_n) \ge 1/(3n)$. Hence the discrepancy method gives at most an $\Omega(\log n)$ lower bound for DISJ_n.*

Proof. For $i \in [n]$, the rectangle $R_i = \{x : x_i = 1\} \times \{y : y_i = 1\}$ is a 0-rectangle, and these n rectangles cover $\mathrm{DISJ}^{-1}(0)$. Let $p_0 = \mu(\mathrm{DISJ}^{-1}(0))$. If $p_0 \ge 1/3$, some R_i has $\mathrm{disc}_\mu(\mathrm{DISJ}, R_i) = \mu(R_i) \ge p_0/n \ge 1/(3n)$. Otherwise the whole space, which is also a rectangle, has discrepancy $|1 - 2p_0| \ge 1/3$. □

Nevertheless $\mathrm{R}^{\mathrm{pub}}(\mathrm{DISJ}_n) = \Omega(n)$, as we will prove in Chapter 10. The table below summarises the picture for the running examples.

	EQ_n	GT_n	IP_n	DISJ_n
D	$n+1$	$\Theta(n)$	$\Theta(n)$	$n+1$
R	$\Theta(\log n)$	$\Theta(\log n)$	$\Theta(n)$	$\Theta(n)$ (Chapter 10)
$\mathrm{R}^{\mathrm{pub}}$	$\Theta(1)$	$\Theta(\log n)$	$\Theta(n)$	$\Theta(n)$ (Chapter 10)

Looking ahead. The two entries for disjointness cannot be obtained with the methods seen so far: discrepancy fails by Proposition 7.7, and Theorem 5.14 only gives $\Omega(\log n)$. The proof in Part III exploits the fact that DISJ_n is built from n independent copies of a tiny one-bit problem. We therefore first study, in Chapter 8, how the complexity of solving many copies of a problem relates to the complexity of solving one; this question leads naturally to measuring *information* rather than bits (Chapters 9 and 10).

7.5 Notes and references

The discrepancy method and the $\Omega(n)$ bound for inner product are due to Chor and Goldreich [CG88]; the matrix-norm bound is often called Lindsey's lemma. See [KN97, Ch. 3] and [RY20, Ch. 5], the latter with many further applications. For sharper results on greater-than see Viola [Vio15]. The limitation in Proposition 7.7 motivates the *corruption* (or rectangle)

bound, with which Razborov [Raz92] proved the linear lower bound for disjointness; we follow the information-theoretic route instead.

Part III

Many Copies and Information

8 Many Copies: Direct Sums and Amortization

Disjointness is the conjunction of n independent copies of a one-bit problem: x and y are disjoint iff for every coordinate i the bits x_i, y_i are not both 1. This suggests a general question. If solving one instance of a problem costs c bits, does solving t independent instances cost about tc bits? This is the *direct-sum question*. For many natural problems the answer is "roughly yes", but there are surprising exceptions, and the general question is still open. This chapter presents simple upper bounds, an example (for relations) where batching helps enormously, and a general lower bound for functions due to Feder, Kushilevitz, Naor and Nisan.

8.1 Definitions

Let $f\colon X \times Y \to Z$ and $t \geq 1$. Alice receives $\boldsymbol{x} = (x_1, \ldots, x_t) \in X^t$, Bob receives $\boldsymbol{y} = (y_1, \ldots, y_t) \in Y^t$, and

$$f^t(\boldsymbol{x}, \boldsymbol{y}) = (f(x_1, y_1), \ldots, f(x_t, y_t)) \in Z^t.$$

For Boolean f we also define

$$f^{\wedge t}(\boldsymbol{x}, \boldsymbol{y}) = \bigwedge_{i=1}^{t} f(x_i, y_i), \qquad f^{\vee t}(\boldsymbol{x}, \boldsymbol{y}) = \bigvee_{i=1}^{t} f(x_i, y_i).$$

For instance, with $\mathrm{AND}(a, b) = ab$ and $\mathrm{NAND} = 1 - \mathrm{AND}$ on single bits,

$$\mathrm{DISJ}_n = \mathrm{NAND}^{\wedge n}, \qquad 1 - \mathrm{DISJ}_n = \mathrm{AND}^{\vee n},$$

a representation that is the starting point of the lower bound in Chapter 10.

8.2 Upper bounds

Proposition 8.1. *For every f and $t \geq 1$:*

1. $\mathrm{D}(f^t) \leq t\,\mathrm{D}(f)$;
2. $\mathrm{R}^{\mathrm{pub}}(f^t) \leq O(t \log t) \cdot \mathrm{R}^{\mathrm{pub}}(f)$ *(for Boolean f);*
3. $\mathrm{R}(f^t) \leq O(t \log t) \cdot \mathrm{R}^{\mathrm{pub}}(f) + O(\log(tn))$ *if $X = Y = \{0,1\}^n$.*

Proof. (1) Solve the copies one after another. (2) Reduce the error of a public-coin protocol for f to $1/(3t)$ by $O(\log t)$ repetitions (Lemma 5.2) and run it on every copy; by the union bound all t answers are correct with probability at least $2/3$. (3) Apply Newman's theorem (Theorem 5.12) to the protocol of (2); the inputs of f^t have tn bits, so the overhead is $O(\log(tn))$. (Newman's theorem does not require Boolean outputs.) □

Example 8.2 *(Many equalities).* Since $\mathrm{R}^{\mathrm{pub}}(\mathrm{EQ}_n) = O(1)$, we get $\mathrm{R}(\mathrm{EQ}_n^t) = O(t \log t + \log n)$. For $t = \log n$ this is $O(\log n \log\log n)$, compared with $t \cdot \mathrm{R}(\mathrm{EQ}_n) = \Theta(\log^2 n)$ for solving the copies separately. Feder et al. [FKNN95] showed that even $O(t + \log n)$ (in expectation) is possible, so the *amortized* randomized complexity of equality is $O(1)$, although $\mathrm{R}(\mathrm{EQ}_n) = \Theta(\log n)$.

For $\wedge$ and $\vee$ the situation is asymmetric.

Proposition 8.3. $\mathrm{R}^{\mathrm{pub}}(\mathrm{EQ}_n^{\wedge t}) = O(1)$ *and* $\mathrm{R}^{\mathrm{pub}}(\mathrm{EQ}_n^{\vee t}) = O(t)$.

Proof. $\mathrm{EQ}_n^{\wedge t}(\boldsymbol{x},\boldsymbol{y}) = \mathrm{EQ}_{tn}(x_1\cdots x_t,\, y_1\cdots y_t)$, so Proposition 5.6 applies directly.

For $\mathrm{EQ}_n^{\vee t}$ we use the one-sided single test of Proposition 5.6 (two bits; it always says "equal" on equal strings and says "equal" with probability exactly $1/2$ on unequal strings). The players keep a pointer i, initially 1, and perform $8t$ rounds: in each round they test the pair (x_i, y_i) with fresh public randomness, and if the test says "unequal" they increment i. Once $i > t$ they stop and output 0; if $i \le t$ after $8t$ rounds they output 1. The cost is at most $16t$.

If $x_j = y_j$ for some j, the pointer can never move past j (the test never says "unequal" on equal strings), so the output 1 is always correct. If $x_j \neq y_j$ for every j, the pointer advances in each round independently with probability $1/2$, and the protocol errs only if fewer than t of the $8t$ rounds advance. By Hoeffding's inequality this has probability at most $e^{-2(3t)^2/(8t)} = e^{-9t/4} < 0.11$. □

Remark 8.4. $O(t)$ is optimal for $\mathrm{EQ}^{\vee t}$: with $x_i = (a_i, 1)$ and $y_i = (1, b_i)$ we get $\mathrm{EQ}_2(x_i, y_i) = a_i b_i$, so $\mathrm{EQ}_2^{\vee t}$ computes $1 - \mathrm{DISJ}_t(a, b)$, which needs $\Omega(t)$ bits by Theorem 10.16.

8.3 Relations: batching can help a lot

A *relation* is a set $F \subseteq X \times Y \times Z$; on input (x, y) the players must output some z with $(x, y, z) \in F$ (we assume one exists). Functions are the special case in which exactly one z is allowed. Many quantities, including D and F^t, are defined for relations exactly as for functions.

Example 8.5. Let n be even. Alice receives $x \in \{0,1\}^n$ with exactly $n/2$ ones; Bob receives nothing. They must output an index i with $x_i = 1$. Call this relation F.

Proposition 8.6. $\mathrm{D}(F) = \Theta(\log n)$. *More precisely* $\log(n/2+1) \le \mathrm{D}(F) \le \lceil \log n \rceil$.

Proof. For the upper bound, Alice sends any index of a 1. For the lower bound, Bob has no input, so the whole transcript is a function of x; with c bits there are at most 2^c transcripts and hence at most 2^c possible outputs. If $2^c \le n/2$, Alice's input could have its $n/2$ ones exactly outside this set of outputs, and the protocol would fail. □

Naively, t copies should cost about $t \log n$ bits. They do not.

Proposition 8.7. $\mathrm{D}(F^t) \le t + \log n + \log t + 2 = O(t + \log n)$.

Proof. *A hitting set.* We claim there is a set $J \subseteq [n]^t$ with $|J| \le nt2^t + 1$ such that for every legal input $(x^{(1)}, \ldots, x^{(t)})$ some $j = (j_1, \ldots, j_t) \in J$ has $x^{(1)}_{j_1} = \cdots = x^{(t)}_{j_t} = 1$. Choose $m = \lceil nt2^t \rceil$ tuples independently and uniformly from $[n]^t$. For a fixed legal input, a random tuple hits a 1 in all t coordinates with probability exactly 2^{-t}, so all m tuples fail with probability $(1-2^{-t})^m \le e^{-m2^{-t}} \le e^{-nt}$. There are at most 2^{nt} legal inputs, and $2^{nt}e^{-nt} < 1$; so some choice of m tuples works for all of them.

The protocol. Fix such a J. Alice sends the index (in J) of a tuple that hits a 1 in every copy, using $\lceil \log |J| \rceil \le t + \log n + \log t + 2$ bits. □

So the amortized cost per copy is $O(1 + \log(n)/t)$, which tends to a constant, while a single copy needs $\Theta(\log n)$. Feder et al. [FKNN95] showed that even partial *functions* can behave this way. For total functions, however, such a dramatic saving is impossible, as we show next.

8.4 A direct-sum lower bound for functions

The bound goes through covers. Recall that $C(f)$ is the minimum number of monochromatic rectangles covering the whole matrix, and that by (3.1) $\mathrm{D}(f) = O(\log^2 C(f))$. For a non-Boolean function such as f^t, "monochromatic" means that the whole output vector is constant on the rectangle.

Lemma 8.8 ([FKNN95]). *Let $f\colon \{0,1\}^n \times \{0,1\}^n \to \{0,1\}$ and $t \geq 1$. Then*

$$C(f) \;\leq\; \left\lceil 2n\, C(f^t)^{1/t} \right\rceil .$$

Proof. Let $K = C(f^t)$ and fix an optimal cover of M_{f^t}. We build a cover of M_f greedily. Suppose $S \subseteq \{0,1\}^n \times \{0,1\}^n$ is the set of entries not covered yet. The set S^t consists of inputs to f^t (a t-tuple of uncovered entries, one per copy), and it is covered by the K rectangles, so one of them, $\mathcal{R} = \mathcal{A} \times \mathcal{B}$, contains at least $|S|^t/K$ elements of S^t.

For $i \in [t]$ let $R_i = A_i \times B_i$, where A_i and B_i are the projections of $\mathcal{A}$ and $\mathcal{B}$ to the i-th copy.

- R_i is an f-monochromatic rectangle: if $x_i \in A_i$ and $y_i \in B_i$, choose $\boldsymbol{x} \in \mathcal{A}$ and $\boldsymbol{y} \in \mathcal{B}$ with these i-th coordinates; then $(\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R}$, and $f(x_i, y_i)$ is the i-th coordinate of the constant value of f^t on $\mathcal{R}$.
- If $(\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{R} \cap S^t$, then $(x_i, y_i) \in R_i \cap S$ for every i. Hence $|S|^t/K \leq |\mathcal{R} \cap S^t| \leq \prod_{i=1}^{t} |R_i \cap S|$.

So some i has $|R_i \cap S| \geq |S|/K^{1/t}$: one monochromatic rectangle of M_f covers at least a $K^{-1/t}$ fraction of the uncovered entries. Starting with 4^n uncovered entries, after m greedy steps at most $4^n(1 - K^{-1/t})^m \leq 4^n e^{-m/K^{1/t}}$ remain, and for $m = \left\lceil 2nK^{1/t} \right\rceil$ this is at most $(4/e^2)^n < 1$. □

Theorem 8.9 (Feder–Kushilevitz–Naor–Nisan [FKNN95]). *There is an absolute constant $\alpha > 0$ such that for every $f\colon \{0,1\}^n \times \{0,1\}^n \to \{0,1\}$ and $t \geq 1$,*

$$\mathrm{D}(f^t) \;\geq\; t\Big(\alpha\sqrt{\mathrm{D}(f)} - \log(4n)\Big).$$

In particular the amortized complexity $\mathrm{D}(f^t)/t$ is at least $\Omega(\sqrt{\mathrm{D}(f)}) - O(\log n)$.

Proof. By (3.1) there is a constant β with $\mathrm{D}(f) \leq \beta \log^2 C(f)$ (when $C(f) \geq 2$; otherwise f is constant and there is nothing to prove). A protocol for f^t of cost $\mathrm{D}(f^t)$ partitions the matrix of f^t into at most $2^{\mathrm{D}(f^t)}$ monochromatic rectangles, so $\log C(f^t) \leq \mathrm{D}(f^t)$. By Lemma 8.8, $C(f) \leq 2nC(f^t)^{1/t} + 1 \leq 4nC(f^t)^{1/t}$, hence

$$\sqrt{\mathrm{D}(f)/\beta} \;\leq\; \log C(f) \;\leq\; \log(4n) + \frac{\log C(f^t)}{t} \;\leq\; \log(4n) + \frac{\mathrm{D}(f^t)}{t}.$$

Rearranging gives the claim with $\alpha = 1/\sqrt{\beta}$. □

Remark 8.10. Theorem 8.9 is weak when $\mathrm{D}(f)$ is small, and it says nothing for relations, as Proposition 8.7 shows it must not. Whether $\mathrm{D}(f^t) = \Omega(t\,\mathrm{D}(f))$ for every function f (the *direct-sum conjecture* for deterministic communication) is open. For randomized protocols, information complexity gives a clean answer in an amortized sense: Braverman and Rao [BR14] proved that the information complexity of f (with respect to an input distribution) equals the amortized communication needed to solve many independent copies of f with small error per copy; see [RY20, Ch. 7].

Looking ahead. The last remark points to the central idea of the remaining chapters. Communication is hard to split among independent copies, but *information* splits cleanly: by the chain rule, the information that a transcript reveals about n independent coordinates is at least the sum of what it reveals about each of them. To make this precise we first develop the basic notions of information theory (Chapter 9), and then use them to prove the linear lower bound for disjointness (Chapter 10).

8.5 Notes and references

The direct-sum question was studied systematically by Feder, Kushilevitz, Naor and Nisan [FKNN95], who proved Lemma 8.8 and Theorem 8.9 and gave examples (for partial functions and for randomized protocols) where amortization helps. See also [RY20, Ch. 1] for a discussion of direct sums in the deterministic setting and [RY20, Ch. 7] for the randomized one.

9 Entropy, Divergence and Mutual Information

The lower bounds of Parts I and II were combinatorial: they bounded the number, the size or the bias of rectangles. The lower bound for set disjointness, the strongest result in these notes, is instead proved by measuring *information*: how much a transcript reveals about the inputs. As explained at the end of Chapter 8, the advantage of information over communication is that it adds up over independent coordinates. This chapter develops the necessary vocabulary from Shannon's information theory [Sha48]. Everything is over finite probability spaces, and the standard reference is Cover and Thomas [CT06].

9.1 Self-information and prefix-free codes

Let π be a distribution on a finite set Ω.

Definition 9.1 (Self-information). The *self-information* of $\omega \in \Omega$ is $I_\pi(\omega) = \log \frac{1}{\pi(\omega)}$ (and $+\infty$ if $\pi(\omega) = 0$).

Rare outcomes carry much information. The following definitions make "$I_\pi(\omega)$ bits" literal.

Definition 9.2 (Prefix-free code). A map $C\colon \Omega \to \{0,1\}^*$ is a *prefix-free code* if for $\omega \neq \omega'$ the string $C(\omega)$ is not a prefix of $C(\omega')$.

Prefix-free codewords can be concatenated and still be decoded unambiguously. They correspond to the leaves of a binary tree. In particular, the transcripts of a protocol tree (the root-to-leaf bit strings) form a prefix-free code.

Lemma 9.3 (Kraft's inequality). *If C is prefix-free, then $\sum_\omega 2^{-|C(\omega)|} \leq 1$. Conversely, if nonnegative integers $(\ell_\omega)_{\omega\in\Omega}$ satisfy $\sum_\omega 2^{-\ell_\omega} \leq 1$, there is a prefix-free code with $|C(\omega)| = \ell_\omega$.*

Proof. Let L be the maximum codeword length. A codeword of length ℓ is a prefix of exactly $2^{L-\ell}$ strings of length L, and by prefix-freeness these sets are disjoint for different codewords. Hence $\sum_\omega 2^{L-|C(\omega)|} \leq 2^L$.

Conversely, order $\Omega = \{\omega_1, \dots, \omega_m\}$ so that $\ell_1 \leq \ell_2 \leq \cdots \leq \ell_m$ (writing $\ell_k = \ell_{\omega_k}$) and choose codewords greedily in this order, $C(\omega_k)$ being any string of length ℓ_k that has none of the previously chosen codewords as a prefix. Each earlier codeword $C(\omega_i)$, $i < k$, excludes exactly $2^{\ell_k - \ell_i}$ strings of length ℓ_k (and no later choice can be a prefix of an earlier one, since lengths are nondecreasing). So the number of available strings is at least

$$2^{\ell_k}\Big(1 - \sum_{i<k} 2^{-\ell_i}\Big) \geq 2^{\ell_k} \sum_{i \geq k} 2^{-\ell_i} \geq 1. \qquad \square$$

Corollary 9.4 (Shannon code). *For every distribution π there is a prefix-free code C_π on* $\operatorname{supp}(\pi)$ *with $|C_\pi(\omega)| = \lceil I_\pi(\omega) \rceil$.*

Proof. $\sum_\omega 2^{-\lceil \log(1/\pi(\omega)) \rceil} \leq \sum_\omega \pi(\omega) = 1$; apply Lemma 9.3. $\square$

9.2 Entropy

Definition 9.5 (Entropy). The *entropy* of π is the expected self-information,

$$H(\pi) = \mathop{\mathbb{E}}_{\omega\sim\pi}[I_\pi(\omega)] = \sum_\omega \pi(\omega) \log \frac{1}{\pi(\omega)},$$

with the convention $0 \log \frac{1}{0} = 0$.

Proposition 9.6. $0 \le H(\pi) \le \log|\operatorname{supp}(\pi)|$, *with equality on the right iff* π *is uniform on its support.*

Proof. Each term is nonnegative. By Jensen's inequality for the concave function log, $H(\pi) = \mathbb{E}[\log \frac{1}{\pi(\omega)}] \le \log \mathbb{E}[\frac{1}{\pi(\omega)}] = \log|\operatorname{supp}(\pi)|$, with equality iff $1/\pi(\omega)$ is constant on the support. □

Theorem 9.7 (Source coding). *For every* π,

$$H(\pi) \;\le\; \min_{C} \mathop{\mathbb{E}}_{\omega\sim\pi} |C(\omega)| \;\le\; H(\pi)+1,$$

the minimum being over prefix-free codes.

Proof. The upper bound: the Shannon code has expected length $\mathbb{E}\lceil I_\pi(\omega)\rceil \le H(\pi)+1$. The lower bound: for any prefix-free C, by Jensen and Kraft,

$$H(\pi) - \mathbb{E}\,|C(\omega)| = \mathbb{E}\Big[\log \frac{2^{-|C(\omega)|}}{\pi(\omega)}\Big] \le \log \mathbb{E}\Big[\frac{2^{-|C(\omega)|}}{\pi(\omega)}\Big] \le \log \sum_{\omega} 2^{-|C(\omega)|} \le 0. \qquad \square$$

So entropy is, up to one bit, the minimum expected number of bits needed to describe a sample of π. For communication complexity the crude consequence of Proposition 9.6 is the most important one: *a transcript of length at most* c *has entropy at most* c.

9.3 Divergence and Pinsker's inequality

Definition 9.8 (Cross-entropy and divergence). Let π, τ be distributions on Ω. The *cross-entropy* is $H(\pi\,\|\,\tau) = \mathbb{E}_{\omega\sim\pi}[I_\tau(\omega)]$ and the *Kullback–Leibler divergence* is

$$D_{\mathrm{KL}}(\pi\,\|\,\tau) = H(\pi\,\|\,\tau) - H(\pi) = \sum_{\omega} \pi(\omega)\log\frac{\pi(\omega)}{\tau(\omega)}$$

(which is $+\infty$ if $\tau(\omega)=0<\pi(\omega)$ for some ω).

$H(\pi\,\|\,\tau)$ is roughly the expected length of the Shannon code of τ when the samples actually come from π, so $D_{\mathrm{KL}}(\pi\,\|\,\tau)$ is the price, in bits, of using the wrong distribution. It is not symmetric and is not a metric.

Proposition 9.9 (Gibbs' inequality). $D_{\mathrm{KL}}(\pi\,\|\,\tau) \ge 0$, *with equality iff* $\pi = \tau$.

Proof. By Jensen's inequality for the strictly concave function log,

$$-D_{\mathrm{KL}}(\pi\,\|\,\tau) = \mathop{\mathbb{E}}_{\omega\sim\pi}\Big[\log\frac{\tau(\omega)}{\pi(\omega)}\Big] \le \log \sum_{\omega\in\operatorname{supp}\pi} \tau(\omega) \le 0.$$

Equality forces τ/π to be constant on $\operatorname{supp}(\pi)$ and $\tau(\operatorname{supp}\pi)=1$, i.e. $\pi=\tau$. □

Lemma 9.10 (Log-sum inequality). *For nonnegative* $a_1,\dots,a_k$ *and positive* $b_1,\dots,b_k$, *with* $A=\sum a_i$ *and* $B=\sum b_i$,

$$\sum_i a_i \log\frac{a_i}{b_i} \ge A\log\frac{A}{B}.$$

Proof. If $A=0$ both sides vanish. Otherwise let $\pi(i)=a_i/A$ and $\tau(i)=b_i/B$. Then the left side equals $A\sum_i \pi(i)\big(\log\frac{\pi(i)}{\tau(i)} + \log\frac{A}{B}\big) = A\,D_{\mathrm{KL}}(\pi\,\|\,\tau) + A\log\frac{A}{B}$, and we use Proposition 9.9. □

A consequence is that divergence can only decrease when outcomes are merged (*data processing*): if Ω is partitioned into blocks $\Omega_1, \dots, \Omega_k$, then $D_{\mathrm{KL}}(\pi \,\|\, \tau) \geq \sum_j \pi(\Omega_j) \log \frac{\pi(\Omega_j)}{\tau(\Omega_j)}$; apply the log-sum inequality within each block.

We measure the distance between distributions by $\|\pi - \tau\|_1 = \sum_\omega |\pi(\omega) - \tau(\omega)|$. It is twice the largest difference in the probability of an event:

$$\|\pi - \tau\|_1 = 2 \max_{E \subseteq \Omega} \big(\pi(E) - \tau(E)\big), \tag{9.1}$$

the maximum being attained at $E = \{\omega : \pi(\omega) \geq \tau(\omega)\}$. In particular $|\pi(E) - \tau(E)| \leq \frac{1}{2} \|\pi - \tau\|_1$ for every event E.

Theorem 9.11 (Pinsker's inequality). $\|\pi - \tau\|_1^2 \leq (2 \ln 2)\, D_{\mathrm{KL}}(\pi \,\|\, \tau)$.

Proof. Let $E = \{\omega : \pi(\omega) \geq \tau(\omega)\}$, $p = \pi(E)$ and $q = \tau(E)$, so that $\|\pi - \tau\|_1 = 2(p - q)$ by (9.1). Merging Ω into the two blocks $E, \bar{E}$ (data processing),

$$D_{\mathrm{KL}}(\pi \,\|\, \tau) \geq p \log \frac{p}{q} + (1 - p) \log \frac{1 - p}{1 - q}.$$

It therefore suffices to show, in natural logarithms, that

$$\phi(q) := p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q} - 2(p - q)^2 \geq 0 \qquad \text{for } q \in (0, 1).$$

We have $\phi(p) = 0$ and

$$\phi'(q) = -\frac{p}{q} + \frac{1 - p}{1 - q} + 4(p - q) = (q - p)\Big(\frac{1}{q(1 - q)} - 4\Big).$$

Since $q(1 - q) \leq 1/4$, the second factor is nonnegative; so ϕ decreases on $(0, p]$ and increases on $[p, 1)$, and is minimised at $q = p$. Hence the divergence in nats is at least $2(p - q)^2 = \frac{1}{2} \|\pi - \tau\|_1^2$, and in bits at least $\frac{1}{2 \ln 2} \|\pi - \tau\|_1^2$. (If $q \in \{0, 1\}$ the divergence is infinite or the claim is trivial.) □

For our purposes the constant is irrelevant; the form $\|\pi - \tau\|_1^2 = O(D_{\mathrm{KL}}(\pi \,\|\, \tau))$ is what will be used. The point is that a small divergence forces a small statistical distance.

9.4 Random variables and conditional entropy

For a random variable X on a finite probability space, π_X denotes its distribution, $\pi_X(x) = \Pr[X = x]$, and $H(X) = H(\pi_X)$. For two random variables, $H(X, Y)$ is the entropy of the pair, i.e. of the joint distribution $\pi_{X,Y}$.

Proposition 9.12 (Subadditivity). *$H(X, Y) \leq H(X) + H(Y)$, with equality iff X and Y are independent. More generally $H(X_1, \dots, X_k) \leq \sum_i H(X_i)$.*

Proof. $H(X) + H(Y) - H(X, Y) = \sum_{x,y} \pi_{X,Y}(x, y) \log \frac{\pi_{X,Y}(x,y)}{\pi_X(x)\pi_Y(y)} = D_{\mathrm{KL}}(\pi_{X,Y} \,\|\, \pi_X \times \pi_Y) \geq 0$, with equality iff $\pi_{X,Y} = \pi_X \times \pi_Y$. The general case follows by induction. □

Definition 9.13 (Conditional entropy). For x with $\Pr[X = x] > 0$ let $\pi_{Y|X=x}$ be the conditional distribution of Y, and $H(Y \mid X = x) = H(\pi_{Y|X=x})$. The *conditional entropy* is $H(Y \mid X) = \mathbb{E}_{x \sim \pi_X} H(Y \mid X = x)$. Conditional versions such as $H(Y \mid X, Z)$ or $H(X, Y \mid Z)$ are defined in the same way, by applying the definitions to the conditional distributions given $Z = z$ and averaging over z.

Proposition 9.14 (Chain rule). *$H(X,Y) = H(X) + H(Y \mid X)$; more generally $H(X_1,\dots,X_k \mid Z) = \sum_{i=1}^{k} H(X_i \mid X_1,\dots,X_{i-1},Z)$.*

Proof. Take expectations of the identity $\log \frac{1}{\pi_{X,Y}(x,y)} = \log \frac{1}{\pi_X(x)} + \log \frac{1}{\pi_{Y\mid X=x}(y)}$. The general statement follows by induction and by conditioning on $Z = z$. □

Corollary 9.15. *1.* Conditioning reduces entropy: *$H(Y \mid X,Z) \le H(Y \mid Z)$.*

2. *$H(X,Y \mid Z) \le H(X \mid Z) + H(Y \mid Z)$.*

3. Functions reduce entropy: *$H(g(X)) \le H(X)$ for every function g.*

Proof. (1) For Z constant: $H(Y \mid X) = H(X,Y) - H(X) \le H(Y)$ by the chain rule and subadditivity; in general apply this to the distributions conditioned on $Z = z$ and average. (2) Chain rule and (1). (3) The pair $(X, g(X))$ carries the same information as X, so $H(X) = H(X,g(X)) = H(g(X)) + H(X \mid g(X)) \ge H(g(X))$. □

Proposition 9.16 (A Shearer-type inequality). *For random variables X,Y,Z,*

$$2H(X,Y,Z) \le H(X,Y) + H(Y,Z) + H(X,Z).$$

Proof. By the chain rule, $H(X,Y,Z) = H(X,Y) + H(Z \mid X,Y)$ and $H(X,Y,Z) = H(Y,Z) + H(X \mid Y,Z)$. By Corollary 9.15, $H(Z \mid X,Y) + H(X \mid Y,Z) \le H(Z \mid X) + H(X) = H(X,Z)$. Add the two identities. □

Example 9.17 *(Loomis–Whitney).* Let $S \subseteq \mathbb{R}^3$ be finite and let (X,Y,Z) be uniform on S. Then $H(X,Y,Z) = \log|S|$, and $H(X,Y) \le \log|S_{xy}|$, where S_{xy} is the projection of S to the xy-plane (Proposition 9.6); similarly for the other two planes. Proposition 9.16 gives

$$|S|^2 \le |S_{xy}| \cdot |S_{yz}| \cdot |S_{xz}|.$$

For example, a set of n^3 points has a projection to some coordinate plane with at least n^2 points. This is the discrete Loomis–Whitney inequality [LW49]; the general statement behind it is Shearer's lemma [CGFS86].

9.5 Mutual information

Definition 9.18 (Mutual information).

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X),$$

$$I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z) = \mathop{\mathbb{E}}_{z}\big[I(X;Y) \text{ conditioned on } Z = z\big].$$

$I(X;Y)$ is the reduction in uncertainty about X obtained by learning Y. By Proposition 9.12 and Corollary 9.15, $0 \le I(X;Y \mid Z) \le H(X \mid Z) \le H(X)$, and $I(X;Y) = 0$ iff X and Y are independent.

Proposition 9.19.

$$I(X;Y) = D_{\mathrm{KL}}(\pi_{X,Y} \,\|\, \pi_X \times \pi_Y) = \mathop{\mathbb{E}}_{y\sim\pi_Y} D_{\mathrm{KL}}\Big(\pi_{X\mid Y=y} \,\Big\|\, \pi_X\Big).$$

Proof. The first identity is the computation in the proof of Proposition 9.12. For the second, write $\pi_{X,Y}(x,y) = \pi_Y(y)\pi_{X\mid Y=y}(x)$:

$$\sum_{x,y} \pi_{X,Y}(x,y) \log \frac{\pi_{X,Y}(x,y)}{\pi_X(x)\pi_Y(y)} = \sum_{y} \pi_Y(y) \sum_{x} \pi_{X\mid Y=y}(x) \log \frac{\pi_{X\mid Y=y}(x)}{\pi_X(x)}.$$ □

The second form says: mutual information is the average divergence between what we believe about X after seeing Y and what we believed before. Combined with Pinsker's inequality, it says that if Y carries little information about X, then the conditional distributions of X given Y are typically close to the prior.

Proposition 9.20 (Chain rule for mutual information).

$$I(X_1, \dots, X_k; Y \mid Z) = \sum_{i=1}^{k} I(X_i; Y \mid X_1, \dots, X_{i-1}, Z).$$

Proof. Write the left side as $H(X_1, \dots, X_k \mid Z) - H(X_1, \dots, X_k \mid Y, Z)$ and expand both terms with the chain rule for entropy. □

Mutual information is symmetric, so the chain rule can also be applied to the second argument. The following three facts will be used in Chapter 10.

Lemma 9.21. *Let M, X, W, V be random variables.*

1. (Information is bounded by length.) $I(M; X \mid V) \le H(M) \le \log |\operatorname{supp}(M)|$.
2. (Superadditivity for independent parts.) *If $X = (X_1, \dots, X_n)$ and the X_i are mutually independent conditioned on every value of V, then $I(M; X \mid V) \ge \sum_{i=1}^{n} I(M; X_i \mid V)$.*
3. (Conditioning on independent variables.) *If $I(X; W \mid V) = 0$, then $I(M; X \mid V, W) \ge I(M; X \mid V)$.*

Proof. (1) $I(M; X \mid V) = H(M \mid V) - H(M \mid X, V) \le H(M \mid V) \le H(M)$.
(2) By the chain rule, $I(M; X \mid V) = \sum_i I(M; X_i \mid X_{<i}, V)$, and

$$\begin{aligned} I(M; X_i \mid X_{<i}, V) &= H(X_i \mid X_{<i}, V) - H(X_i \mid M, X_{<i}, V) \\ &\ge H(X_i \mid V) - H(X_i \mid M, V) = I(M; X_i \mid V), \end{aligned}$$

where we used $H(X_i \mid X_{<i}, V) = H(X_i \mid V)$ (independence) and that conditioning reduces entropy.
(3) Expand $I(M, W; X \mid V)$ in two ways: $I(W; X \mid V) + I(M; X \mid W, V) = I(M; X \mid V) + I(W; X \mid M, V)$. The first term on the left is 0 and the last term on the right is nonnegative. □

Part (2) can fail without independence, and part (3) can fail when W is correlated with X: conditioning can increase or decrease mutual information.

Definition 9.22 (Jensen–Shannon divergence). For distributions π, τ on Ω, with $\bar{\pi} = \frac{1}{2}(\pi + \tau)$,

$$\mathrm{JS}(\pi, \tau) = \tfrac{1}{2} D_{\mathrm{KL}}(\pi \,\|\, \bar{\pi}) + \tfrac{1}{2} D_{\mathrm{KL}}(\tau \,\|\, \bar{\pi}).$$

Unlike $D_{\mathrm{KL}}(\cdot \,\|\, \cdot)$, the JS divergence is symmetric and always finite (at most 1).

Proposition 9.23. *Let B be uniform on $\{0, 1\}$ and let X be any random variable. Then*

$$I(X; B) = \mathrm{JS}(\pi_{X|B=0}, \pi_{X|B=1}).$$

Proof. By Proposition 9.19 (with the roles of the variables chosen suitably), $I(X; B) = \frac{1}{2} D_{\mathrm{KL}}\big(\pi_{X|B=0} \,\big\|\, \pi_X\big) + \frac{1}{2} D_{\mathrm{KL}}\big(\pi_{X|B=1} \,\big\|\, \pi_X\big)$, and $\pi_X = \frac{1}{2}\pi_{X|B=0} + \frac{1}{2}\pi_{X|B=1}$ because B is uniform. □

9.6 Notes and references

Entropy and the source coding theorem are due to Shannon [Sha48]; Kraft's inequality, Gibbs' inequality, the log-sum inequality and Pinsker's inequality are all in Cover and Thomas [CT06]. For the use of information theory in communication complexity, see [RY20, Ch. 6], whose presentation these notes follow closely in spirit. The Jensen–Shannon divergence was introduced by Lin [Lin91]. Shearer's lemma is from [CGFS86].

10 Information Complexity and the Hardness of Disjointness

We now use the tools of Chapter 9 to prove two lower bounds that the rectangle methods cannot give easily. The first, for the one-way index problem, is a warm-up that shows the basic pattern: *a short message carries little information, and a message carrying little information about x_j cannot help to guess x_j*. The second is the central result of these notes,

$$\mathrm{R}^{\mathrm{pub}}(\mathrm{DISJ}_n) = \Omega(n),$$

which says that randomness does not help for disjointness. Recall from Proposition 7.7 that discrepancy cannot prove this.

10.1 Hellinger distance

The disjointness proof compares transcript distributions, and the most convenient way to do so is the Hellinger distance.

Definition 10.1 (Hellinger distance). For distributions π, τ on a finite set Ω,

$$h(\pi,\tau) = \frac{1}{\sqrt{2}}\big\|\sqrt{\pi}-\sqrt{\tau}\big\|_2 = \Big(\frac{1}{2}\sum_{\omega}\big(\sqrt{\pi(\omega)}-\sqrt{\tau(\omega)}\big)^2\Big)^{1/2},$$

where $\sqrt{\pi}$ denotes the vector $(\sqrt{\pi(\omega)})_\omega$. The *Bhattacharyya coefficient* is $\mathrm{BC}(\pi,\tau) = \sum_\omega \sqrt{\pi(\omega)\tau(\omega)}$.

Proposition 10.2 (Properties of the Hellinger distance). *For all distributions π, τ, ι on Ω:*

1. $h^2(\pi,\tau) = 1 - \mathrm{BC}(\pi,\tau)$*; in particular* $0 \le h \le 1$.
2. (Triangle inequality) $h(\pi,\tau) \le h(\pi,\iota) + h(\iota,\tau)$*; consequently* $h^2(\pi,\tau) \le 2h^2(\pi,\iota) + 2h^2(\iota,\tau)$.
3. (Hellinger vs. ℓ_1) $\|\pi-\tau\|_1 \le 2\sqrt{2}\,h(\pi,\tau)$*, i.e.* $h^2(\pi,\tau) \ge \frac{1}{8}\|\pi-\tau\|_1^2$.
4. (Hellinger vs. divergences) $D_{\mathrm{KL}}(\pi\,\|\,\tau) \ge 2h^2(\pi,\tau)$ *and* $\mathrm{JS}(\pi,\tau) \ge \frac{1}{2}h^2(\pi,\tau)$.

Proof. (1) Expand the square: $\frac{1}{2}\sum(\pi+\tau-2\sqrt{\pi\tau}) = 1-\mathrm{BC}$.

(2) h is $1/\sqrt{2}$ times the Euclidean distance between the vectors $\sqrt{\pi}$ and $\sqrt{\tau}$, hence a metric. The second form follows from $(a+b)^2 \le 2a^2+2b^2$.

(3) By Cauchy–Schwarz,

$$\begin{aligned}\sum_\omega |\pi-\tau| &= \sum_\omega |\sqrt{\pi}-\sqrt{\tau}|(\sqrt{\pi}+\sqrt{\tau}) \\ &\le \Big(\sum(\sqrt{\pi}-\sqrt{\tau})^2\Big)^{1/2}\Big(\sum(\sqrt{\pi}+\sqrt{\tau})^2\Big)^{1/2} \le \sqrt{2h^2}\cdot 2.\end{aligned}$$

(Here $\sum(\sqrt{\pi}+\sqrt{\tau})^2 = 2+2\mathrm{BC} \le 4$.)

(4) In natural logarithms, by Jensen's inequality for the concave ln,

$$\sum_\omega \pi \ln\frac{\pi}{\tau} = -2\sum_{\omega\in\operatorname{supp}\pi}\pi\ln\sqrt{\frac{\tau}{\pi}} \ge -2\ln\sum_{\omega\in\operatorname{supp}\pi}\sqrt{\pi\tau} = -2\ln\mathrm{BC} \ge 2(1-\mathrm{BC}) = 2h^2,$$

using $-\ln u \ge 1-u$. In bits the divergence is larger by the factor $1/\ln 2 > 1$, so $D_{\mathrm{KL}}(\pi\,\|\,\tau) \ge 2h^2(\pi,\tau)$ as well. For the JS divergence, with $\bar\pi = \frac{1}{2}(\pi+\tau)$,

$$\mathrm{JS}(\pi,\tau) \ge h^2(\pi,\bar\pi) + h^2(\tau,\bar\pi) \ge \tfrac{1}{2}\big(h(\pi,\bar\pi)+h(\bar\pi,\tau)\big)^2 \ge \tfrac{1}{2}h^2(\pi,\tau),$$

by the first part of (4), the inequality $a^2+b^2 \ge \frac{1}{2}(a+b)^2$, and the triangle inequality. □

Remark 10.3 *(h^2 is not a metric).* It is tempting to use the triangle inequality in the form $h^2(\pi,\tau) \le h^2(\pi,\iota) + h^2(\iota,\tau)$, but this is false in general: for two point masses $\delta_a \neq \delta_b$ and $\iota = \frac{1}{2}(\delta_a + \delta_b)$ one has $h^2(\delta_a,\delta_b) = 1$ but $h^2(\delta_a,\iota) + h^2(\iota,\delta_b) = 2(1 - 1/\sqrt{2}) \approx 0.59$. The triangle inequality holds for h itself; for h^2 one loses a factor 2, as in (2), which is harmless for our purposes.

10.2 Warm-up: the one-way index problem

Definition 10.4 (Index). In INDEX_n, Alice holds $x \in \{0,1\}^n$ and Bob holds $j \in [n]$; the answer is x_j. In the *one-way* model only Alice may send a message, after which Bob outputs the answer.

With two-way communication the problem is trivial: Bob sends j ($\lceil \log n \rceil$ bits) and Alice replies with x_j. One-way, Alice does not know which bit Bob needs.

Theorem 10.5. *Every one-way public-coin protocol for* INDEX_n *with error* $\varepsilon < 1/2$ *sends at least* $\frac{(1-2\varepsilon)^2}{2\ln 2}\, n$ *bits.*

Proof. By Yao's principle for one-way protocols (see the remark after Theorem 6.5), it suffices to consider a deterministic one-way protocol with error at most ε when X is uniform on $\{0,1\}^n$ and J is uniform on $[n]$, independent of X. Alice sends $M = M(X) \in \{0,1\}^\ell$ and Bob outputs $\beta(J,M)$.

Independence. M is a function of X, and J is independent of X; hence J is independent of the pair (X,M). (Note that X itself is of course *not* independent of M — that is the whole point.) Therefore, for all j and m,

$$\begin{aligned}\Pr[X_J \neq \beta(J,M) \mid J=j, M=m] &= \Pr[X_j \neq \beta(j,m) \mid M=m] \\ &\ge \min_{b\in\{0,1\}} \Pr[X_j = b \mid M=m] = \tfrac{1}{2} - \left|p_{j,m} - \tfrac{1}{2}\right|,\end{aligned}$$

where $p_{j,m} = \Pr[X_j = 0 \mid M = m]$, and J and M are independent as well.

From advantage to information. Since X_j is a uniform bit, $|p_{j,m} - \frac{1}{2}| = \frac{1}{2}\left\|\pi_{X_j|M=m} - \pi_{X_j}\right\|_1$. By Pinsker's inequality (Theorem 9.11), Jensen's inequality for the concave square root, and Proposition 9.19,

$$\begin{aligned}\mathop{\mathbb{E}}_{J,M}\left|p_{J,M} - \tfrac{1}{2}\right| &\le \tfrac{1}{2}\mathop{\mathbb{E}}_{J}\mathop{\mathbb{E}}_{M}\sqrt{2\ln 2 \cdot D_{\mathrm{KL}}\left(\pi_{X_J|M} \,\middle\|\, \pi_{X_J}\right)} \le \tfrac{1}{2}\sqrt{2\ln 2 \cdot \mathop{\mathbb{E}}_{J}\mathop{\mathbb{E}}_{M} D_{\mathrm{KL}}\left(\pi_{X_J|M} \,\middle\|\, \pi_{X_J}\right)} \\ &= \tfrac{1}{2}\sqrt{2\ln 2 \cdot \frac{1}{n}\sum_{j=1}^{n} I(X_j;M)}.\end{aligned}$$

From information to length. The bits $X_1,\dots,X_n$ are independent, so by Lemma 9.21, $\sum_j I(X_j;M) \le I(X;M) \le H(M) \le \ell$.

Putting everything together, $\varepsilon \ge \Pr[X_J \neq \beta(J,M)] \ge \frac{1}{2} - \frac{1}{2}\sqrt{2\ln 2 \cdot \ell/n}$, which rearranges to $\ell \ge (1-2\varepsilon)^2 n/(2\ln 2)$. □

10.3 Information cost

For the two-way setting we measure how much the transcript reveals to each player about the other player's input.

Definition 10.6 (Information cost). Let Π be a private-coin protocol with inputs in $X \times Y$, and let μ be a distribution of a triple (X,Y,Z), where Z is an auxiliary random variable that is

not given to the players. Let M denote the transcript of Π on input (X, Y) (the sequence of all bits sent; it determines the leaf reached). The (internal, conditional) *information cost* of Π is

$$\mathrm{IC}_\mu(\Pi) = I(M; X \mid Y, Z) + I(M; Y \mid X, Z).$$

The first term is what Bob, who knows Y (and, for the analysis, also Z), learns about X from the conversation; the second is what Alice learns about Y. One can define the information complexity of a *function* by optimising over protocols and distributions; for a lower bound, however, we only need one well-chosen μ.

Lemma 10.7 (Information is at most communication). *If Π has cost c, then $\mathrm{IC}_\mu(\Pi) \le 2c$ for every μ.*

Proof. M takes at most 2^c values (the leaves of the tree), so $H(M) \le c$. By Lemma 9.21(1), each of the two terms is at most $H(M)$. □

The proof of the disjointness lower bound follows four steps:

1. reduce public coins to private coins (Newman's theorem);
2. by Lemma 10.7, it suffices to show $\mathrm{IC}_\mu(\Pi) = \Omega(n)$ for every private-coin protocol Π for DISJ_n and a suitable μ;
3. *direct sum*: split $\mathrm{IC}_\mu(\Pi)$ into n terms, one per coordinate, each of which is the information cost of a protocol for the one-bit function NAND;
4. *one-bit bound*: every protocol for NAND has information cost $\Omega(1)$ under the one-bit distribution.

10.4 The hard distribution

Recall $\mathrm{DISJ}_n = \mathrm{NAND}^{\wedge n}$, where $\mathrm{NAND}(a, b) = 1 - ab$.

Definition 10.8 (Hard distribution). Let ν be the uniform distribution on the four triples

$$(A, B, D) \in \{(0,0,0),\ (0,1,0),\ (0,0,1),\ (1,0,1)\}.$$

Equivalently: D is a uniform bit; if $D = 0$ then $A = 0$ and B is a uniform bit; if $D = 1$ then $B = 0$ and A is a uniform bit. Let $\mu = \nu^{\otimes n}$ be the distribution of (X, Y, Z) in which the triples (X_i, Y_i, Z_i), $i \in [n]$, are independent copies of (A, B, D).

Three properties of ν are used below.

- $(A, B) \neq (1, 1)$ always, so $\mathrm{NAND}(A, B) = 1$. Hence μ is supported on 1-inputs of DISJ_n.
- A and B are independent conditioned on D (one of them is constant).
- Under μ, all $3n$ variables are "as independent as possible": conditioned on Z, the pairs (X_i, Y_i) are independent across i, and inside each pair X_i and Y_i are independent.

Remark 10.9. It may look strange that the hard distribution never produces an intersecting pair. The lower bound will not come from distinguishing 0-inputs from 1-inputs inside the distribution. It comes from the fact that the protocol must also be correct on the input $(1, 1)$ in each coordinate, together with the rectangle ("cut-and-paste") structure of private-coin transcripts, which ties the transcript on $(1, 1)$ to the transcripts on $(0, 1)$ and $(1, 0)$.

10.5 The direct-sum step

Lemma 10.10 (Direct sum). *Let Π be a private-coin protocol computing DISJ_n with error at most ε on every input. For every $i \in [n]$ and $z \in \{0,1\}^{[n]\setminus\{i\}}$ there is a private-coin protocol $\Pi_{i,z}$ computing NAND with error at most ε on every input, such that*

$$\mathrm{IC}_\mu(\Pi) \;\geq\; \sum_{i=1}^{n} \mathop{\mathbb{E}}_{z}\big[\mathrm{IC}_\nu(\Pi_{i,z})\big],$$

where z is distributed as $Z_{-i} = (Z_j)_{j\neq i}$, i.e. uniformly.

Proof. Step 1: splitting the information. Conditioned on (Y, Z) the bits $X_1, \ldots, X_n$ are independent, so by Lemma 9.21(2),

$$I(M; X \mid Y, Z) \geq \sum_{i=1}^{n} I(M; X_i \mid Y, Z).$$

Moreover X_i is independent of Y_{-i} conditioned on (Y_i, Z), so by Lemma 9.21(3) (with $V = (Y_i, Z)$ and $W = Y_{-i}$), $I(M; X_i \mid Y, Z) \geq I(M; X_i \mid Y_i, Z)$. The same argument for the other term gives

$$\begin{aligned}\mathrm{IC}_\mu(\Pi) &\geq \sum_{i=1}^{n}\Big(I(M; X_i \mid Y_i, Z) + I(M; Y_i \mid X_i, Z)\Big)\\ &= \sum_{i=1}^{n} \mathop{\mathbb{E}}_{z}\Big[\big(I(M; X_i \mid Y_i, Z_i) + I(M; Y_i \mid X_i, Z_i)\big)\Big|_{Z_{-i}=z}\Big].\end{aligned}$$

Step 2: embedding a one-bit instance. Fix i and z. The protocol $\Pi_{i,z}$ on input $(a, b) \in \{0,1\}^2$ works as follows. Alice sets $X_i = a$, and for $j \neq i$ she sets $X_j = 0$ if $z_j = 0$ and draws X_j uniformly with her private coins if $z_j = 1$. Bob sets $Y_i = b$, and for $j \neq i$ he draws Y_j uniformly with his private coins if $z_j = 0$ and sets $Y_j = 0$ if $z_j = 1$. Then they run Π on (X, Y).

- *It is a private-coin protocol computing* NAND *with error at most ε.* For $j \neq i$ we never have $X_j = Y_j = 1$, so $\mathrm{DISJ}_n(X, Y) = \mathrm{NAND}(a, b)$. Since Π errs with probability at most ε on every input (X, Y), it errs with probability at most ε on average over the sampled coordinates.
- *It has the right information cost.* If $(a, b, d) \sim \nu$, then the joint distribution of (X, Y, M) produced by $\Pi_{i,z}$, together with $Z_i := d$, is exactly the distribution of (X, Y, M, Z_i) under μ conditioned on $Z_{-i} = z$. In particular

$$\begin{aligned}\big(I(M; X_i \mid Y_i, Z_i) + I(M; Y_i \mid X_i, Z_i)\big)\Big|_{Z_{-i}=z} &= I(M; A \mid B, D) + I(M; B \mid A, D)\\ &= \mathrm{IC}_\nu(\Pi_{i,z}),\end{aligned}$$

  where on the right M is the transcript of $\Pi_{i,z}$ on (A, B).

Combining the two steps proves the lemma. □

Remark 10.11 *(Another decomposition).* One can also expand the information cost with the chain rule as

$$\mathrm{IC}_\mu(\Pi) = \sum_{i}\Big(I(M; X_i \mid X_{<i}, Y, Z) + I(M; Y_i \mid X, Y_{>i}, Z)\Big),$$

which is an exact identity. That decomposition is the one needed for general product distributions (it is used in the direct-sum theorems for information complexity [BR14]), where the players must sample $X_{<i}$ and $Y_{>i}$ with *public* randomness. For our distribution, conditioning on Z already makes all coordinates and both sides independent, which is why the simpler inequality of Step 1 suffices and the embedding needs only private coins.

10.6 The one-bit lower bound

Let Π' be a private-coin protocol for NAND on one bit each, and for $(a,b) \in \{0,1\}^2$ let τ_{ab} be the distribution of its transcript on input (a,b).

Lemma 10.12 (Information cost via transcripts). $\mathrm{IC}_\nu(\Pi') = \frac{1}{2}\mathrm{JS}(\tau_{00},\tau_{10}) + \frac{1}{2}\mathrm{JS}(\tau_{00},\tau_{01})$.

Proof. The protocol does not see D, so conditioned on $(A,B) = (a,b)$ (and any value of D) the transcript has distribution τ_{ab}. Consider $I(M;A \mid B,D)$. If $D=0$ then $A=0$ is constant and the information is 0. If $D=1$ then $B=0$ and A is a uniform bit, so by Proposition 9.23 the information is $\mathrm{JS}(\tau_{00},\tau_{10})$. Averaging over D, $I(M;A \mid B,D) = \frac{1}{2}\mathrm{JS}(\tau_{00},\tau_{10})$. Symmetrically $I(M;B \mid A,D) = \frac{1}{2}\mathrm{JS}(\tau_{00},\tau_{01})$. □

Lemma 10.13 (Cut and paste). *For every transcript m, $\tau_{00}(m)\,\tau_{11}(m) = \tau_{01}(m)\,\tau_{10}(m)$. Consequently $h(\tau_{00},\tau_{11}) = h(\tau_{01},\tau_{10})$.*

Proof. A transcript m determines a leaf, and by (5.1) the probability of reaching it on input (a,b) factorises as $\tau_{ab}(m) = p_m(a)q_m(b)$. Hence both products equal $p_m(0)p_m(1)q_m(0)q_m(1)$. Taking square roots and summing over m gives $\mathrm{BC}(\tau_{00},\tau_{11}) = \mathrm{BC}(\tau_{01},\tau_{10})$, and the claim follows from Proposition 10.2(1). □

Lemma 10.14 (Correctness separates $(0,0)$ from $(1,1)$). *If Π' computes* NAND *with error at most ε, then $\|\tau_{00} - \tau_{11}\|_1 \geq 2(1-2\varepsilon)$.*

Proof. The output is a function of the transcript. Let E be the set of transcripts with output 1. Since $\mathrm{NAND}(0,0) = 1$ and $\mathrm{NAND}(1,1) = 0$, $\tau_{00}(E) \geq 1-\varepsilon$ and $\tau_{11}(E) \leq \varepsilon$. Apply (9.1). □

Theorem 10.15 (One-bit information lower bound). *Every private-coin protocol Π' computing* NAND *with error at most ε satisfies $\mathrm{IC}_\nu(\Pi') \geq (1-2\varepsilon)^2/16$.*

Proof. Using, in order, Lemma 10.12, Proposition 10.2(4), the inequality $a^2+b^2 \geq \frac{1}{2}(a+b)^2$, the triangle inequality, Lemma 10.13, Proposition 10.2(3) and Lemma 10.14:

$$\begin{aligned}
\mathrm{IC}_\nu(\Pi') &= \tfrac{1}{2}\mathrm{JS}(\tau_{00},\tau_{10}) + \tfrac{1}{2}\mathrm{JS}(\tau_{00},\tau_{01}) \geq \tfrac{1}{4}\big(h^2(\tau_{10},\tau_{00}) + h^2(\tau_{00},\tau_{01})\big) \\
&\geq \tfrac{1}{8}\big(h(\tau_{10},\tau_{00}) + h(\tau_{00},\tau_{01})\big)^2 \geq \tfrac{1}{8}h^2(\tau_{10},\tau_{01}) = \tfrac{1}{8}h^2(\tau_{00},\tau_{11}) \\
&\geq \tfrac{1}{64}\,\|\tau_{00}-\tau_{11}\|_1^2 \geq \tfrac{1}{64}\cdot 4(1-2\varepsilon)^2 = \tfrac{(1-2\varepsilon)^2}{16}.
\end{aligned}$$

□

The proof is worth reading backwards: the protocol must distinguish $(0,0)$ from $(1,1)$; by cut-and-paste it must then also distinguish $(0,1)$ from $(1,0)$; by the triangle inequality it must distinguish $(0,0)$ from $(0,1)$ or from $(1,0)$; and distinguishing those is exactly what reveals information under ν.

10.7 Disjointness

Theorem 10.16 ([KS92, Raz92, BYJKS04]). *For every $0 \leq \varepsilon < 1/2$,*

$$\mathrm{R}_\varepsilon(\mathrm{DISJ}_n) \geq \frac{(1-2\varepsilon)^2}{32}\,n.$$

Moreover $\mathrm{R}^{\mathrm{pub}}_\varepsilon(\mathrm{DISJ}_n) = \Omega(n)$ for every constant $\varepsilon < 1/2$. Hence $\mathrm{R}(\mathrm{DISJ}_n) = \Theta(n)$ and $\mathrm{R}^{\mathrm{pub}}(\mathrm{DISJ}_n) = \Theta(n)$.

Proof. Let Π be a private-coin protocol of cost c computing DISJ_n with error at most ε. By Lemma 10.7, Lemma 10.10 and Theorem 10.15,

$$2c \geq \mathrm{IC}_\mu(\Pi) \geq \sum_{i=1}^{n} \mathop{\mathbb{E}}_{z} \mathrm{IC}_\nu(\Pi_{i,z}) \geq n \cdot \frac{(1-2\varepsilon)^2}{16}.$$

For public coins, let $\delta = \frac{1}{2}(\frac{1}{2} - \varepsilon)$. By Newman's theorem (Theorem 5.12), $\mathrm{R}^{\mathrm{pub}}_{\varepsilon}(\mathrm{DISJ}_n) \geq \mathrm{R}_{\varepsilon+\delta}(\mathrm{DISJ}_n) - \log n - 2\log(1/\delta) - O(1) \geq \frac{(1-2\varepsilon)^2}{128} n - O(\log n)$. □

Remark 10.17.

1. The first proofs of Theorem 10.16 were combinatorial: Kalyanasundaram and Schnitger [KS92], simplified by Razborov [Raz92] using the corruption bound. The information-theoretic proof presented here is due to Bar-Yossef, Jayram, Kumar and Sivakumar [BYJKS04], building on the notion of information complexity of Chakrabarti, Shi, Wirth and Yao [CSWY01].
2. The constant can be determined exactly: Braverman, Garg, Pankratov and Weinstein [BGPW13] showed that the randomized communication complexity of DISJ_n with vanishing error is $(c + o(1))n$ for an explicit constant $c \approx 0.4827$.
3. Disjointness is the most widely used "hard problem" in applications: lower bounds for streaming algorithms, data structures, property testing, extended formulations and distributed computing are frequently proved by a reduction from it; see [RY20, Part II].

10.8 The big picture

The table collects the complexities of the four running examples established in these notes (all bounds up to constant factors, except where an exact value is given).

	D	R	$\mathrm{R}^{\mathrm{pub}}$	N^1	N^0
EQ_n	$n+1$	$\Theta(\log n)$	$\Theta(1)$	$\Theta(n)$	$\Theta(\log n)$
GT_n	$\Theta(n)$	$\Theta(\log n)$	$\Theta(\log n)$	$\Theta(n)$	$\Theta(n)$
IP_n	$\Theta(n)$	$\Theta(n)$	$\Theta(n)$	$\Theta(n)$	$\Theta(n)$
DISJ_n	$n+1$	$\Theta(n)$	$\Theta(n)$	$\Theta(n)$	$\Theta(\log n)$

(The nondeterministic entries for GT and IP follow from the fooling-set and rectangle-size arguments of Chapter 2: the sets $\{(t, t-1)\}$ and $\{(t,t)\}$ are fooling sets for the values 1 and 0 of GT_n, and every monochromatic rectangle of IP_n has at most 2^n entries while each value is taken on about 2^{2n-1} entries. For $\mathrm{N}^0(\mathrm{EQ}_n) \geq \log n$: if 0-rectangles $A_1 \times B_1, \ldots, A_k \times B_k$ cover all pairs $x \neq y$, the sets $\{i : x \in A_i\}$ must be distinct for distinct x, since $A_i \cap B_i = \varnothing$; so $2^k \geq 2^n$.)

10.9 Notes and references

The one-way index lower bound is due to Ablayev [Abl96] and Kremer, Nisan and Ron [KNR99]; the information-theoretic proof is the standard modern one, see [RY20, Ch. 6]. The Hellinger-distance approach, including the cut-and-paste lemma, is from Bar-Yossef, Jayram, Kumar and Sivakumar [BYJKS04]. Information complexity was introduced in [CSWY01]; its connection to amortized communication is the theorem of Braverman and Rao [BR14], and [RY20, Ch. 7] treats compression of protocols. For the historical lower bounds for disjointness see Kalyanasundaram–Schnitger [KS92] and Razborov [Raz92].

A Probabilistic and Analytic Tools

This appendix collects the elementary inequalities used throughout the notes.

A.1 Elementary inequalities

- *Union bound.* $\Pr[\bigcup_i E_i] \le \sum_i \Pr[E_i]$.
- *Exponential bound.* $1 - u \le e^{-u}$ for all real u; hence $(1-p)^m \le e^{-pm}$ for $p \in [0,1]$.
- *Binomial coefficients.* $\binom{m}{s} \le \left(\frac{em}{s}\right)^s$, and $\sum_{i=0}^{k} \binom{h}{i} \le (h+1)^k$ for integers $0 \le k \le h$.
- *Jensen's inequality.* For concave ϕ (such as log or $\sqrt{\cdot}$) and any random variable V, $\mathbb{E}\,\phi(V) \le \phi(\mathbb{E}\,V)$; the inequality is reversed for convex ϕ. If ϕ is strictly concave, equality holds only if V is almost surely constant.
- *Cauchy–Schwarz.* $|\langle u, v\rangle| \le \|u\|\,\|v\|$; in probabilistic form, $|\,\mathbb{E}[UV]| \le \sqrt{\mathbb{E}\,U^2}\sqrt{\mathbb{E}\,V^2}$.
- *Operator norm.* For a real matrix H, $\|H\| = \max_{\|v\|=1} \|Hv\|$ equals the largest singular value of H, and $|u^\top H v| \le \|u\|\,\|H\|\,\|v\|$. If $HH^\top = sI$ then $\|H\| = \sqrt{s}$.

A.2 Concentration

Lemma A.1 (Hoeffding's inequality [Hoe63]). *Let $V_1, \dots, V_k$ be independent random variables with values in $[0,1]$ and $S = \sum_i V_i$. For every $s \ge 0$,*

$$\Pr[S - \mathbb{E}\,S \ge s] \le e^{-2s^2/k} \qquad \text{and} \qquad \Pr[S - \mathbb{E}\,S \le -s] \le e^{-2s^2/k}.$$

In the analysis of the noisy binary search (Theorem 5.11) the steps are not independent: whether a step is correct depends on where the walk currently is. What we do know is that each step is correct with probability at least p *whatever happened before*. This suffices.

Lemma A.2. *Let $B_1, \dots, B_T \in \{0,1\}$ be random variables such that for every t, conditioned on any history of the process up to time $t-1$, $\Pr[B_t = 1] \ge p$. Then for every $\lambda \ge 0$,*

$$\Pr\Big[\sum_{t=1}^{T} B_t \le (p - \lambda)T\Big] \le e^{-2\lambda^2 T}.$$

Proof sketch. Let $q_t \ge p$ denote the conditional probability that $B_t = 1$ given the history. We may generate B_t as $\mathbf{1}[U_t \le q_t]$, where $U_1, \dots, U_T$ are independent uniform random variables on $[0,1]$, each independent of the history before it. Then $B'_t = \mathbf{1}[U_t \le p] \le B_t$, and the B'_t are independent Bernoulli(p) variables. So $\sum_t B_t \ge \sum_t B'_t$, and Lemma A.1 applied to the B'_t gives the bound. (This is also a special case of the Azuma–Hoeffding inequality for martingales.) □

Bibliography

[Abl96] Farid Ablayev. Lower bounds for one-way probabilistic communication complexity and their application to space complexity. *Theoretical Computer Science*, 157(2):139–159, 1996.

[AKK16] Andris Ambainis, Martins Kokainis, and Robin Kothari. Nearly optimal separations between communication (or query) complexity and partitions. In *Proceedings of the 31st Conference on Computational Complexity (CCC)*, volume 50 of *LIPIcs*, pages 4:1–4:14, 2016.

[AUY83] Alfred V. Aho, Jeffrey D. Ullman, and Mihalis Yannakakis. On notions of information transfer in VLSI circuits. In *Proceedings of the 15th Annual ACM Symposium on Theory of Computing (STOC)*, pages 133–139, 1983.

[BGPW13] Mark Braverman, Ankit Garg, Denis Pankratov, and Omri Weinstein. From information to exact communication. In *Proceedings of the 45th Annual ACM Symposium on Theory of Computing (STOC)*, 2013.

[BR14] Mark Braverman and Anup Rao. Information equals amortized communication. *IEEE Transactions on Information Theory*, 60(10):6058–6069, 2014. Preliminary version in FOCS 2011.

[BYJKS04] Ziv Bar-Yossef, T. S. Jayram, Ravi Kumar, and D. Sivakumar. An information statistics approach to data stream and communication complexity. *Journal of Computer and System Sciences*, 68(4):702–732, 2004.

[CG88] Benny Chor and Oded Goldreich. Unbiased bits from sources of weak randomness and probabilistic communication complexity. *SIAM Journal on Computing*, 17(2):230–261, 1988.

[CGFS86] Fan R. K. Chung, Ronald L. Graham, Peter Frankl, and James B. Shearer. Some intersection theorems for ordered sets and graphs. *Journal of Combinatorial Theory, Series A*, 43(1):23–37, 1986.

[CSWY01] Amit Chakrabarti, Yaoyun Shi, Anthony Wirth, and Andrew Chi-Chih Yao. Informational complexity and the direct sum problem for simultaneous message complexity. In *Proceedings of the 42nd IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 270–278, 2001.

[CT06] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, second edition, 2006.

[FKNN95] Tomás Feder, Eyal Kushilevitz, Moni Naor, and Noam Nisan. Amortized communication complexity. *SIAM Journal on Computing*, 24(4):736–750, 1995. Preliminary version in FOCS 1991.

[FMP+15] Samuel Fiorini, Serge Massar, Sebastian Pokutta, Hans Raj Tiwary, and Ronald de Wolf. Exponential lower bounds for polytopes in combinatorial optimization. *Journal of the ACM*, 62(2):17:1–17:23, 2015.

[FRPU94] Uriel Feige, Prabhakar Raghavan, David Peleg, and Eli Upfal. Computing with noisy information. *SIAM Journal on Computing*, 23(5):1001–1018, 1994.

[GL14] Dmitry Gavinsky and Shachar Lovett. En route to the log-rank conjecture: New reductions and equivalent formulations. In *Proceedings of the 41st International Colloquium on Automata, Languages, and Programming (ICALP)*, volume 8572 of *Lecture Notes in Computer Science*, pages 514–524, 2014.

[GPW15] Mika Göös, Toniann Pitassi, and Thomas Watson. Deterministic communication vs. partition number. In *Proceedings of the 56th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 1077–1088, 2015.

[Hoe63] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[KN97] Eyal Kushilevitz and Noam Nisan. *Communication Complexity*. Cambridge University Press, 1997.

[KNR99] Ilan Kremer, Noam Nisan, and Dana Ron. On randomized one-round communication complexity. *Computational Complexity*, 8(1):21–49, 1999.

[KS92] Bala Kalyanasundaram and Georg Schnitger. The probabilistic communication complexity of set intersection. *SIAM Journal on Discrete Mathematics*, 5(4):545–557, 1992.

[Lin91] Jianhua Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.

[Lov90] László Lovász. Communication complexity: A survey. In Bernhard Korte, László Lovász, Hans Jürgen Prömel, and Alexander Schrijver, editors, *Paths, Flows, and VLSI-Layout*, pages 235–265. Springer, 1990.

[Lov14] Shachar Lovett. Recent advances on the log-rank conjecture in communication complexity. arXiv:1403.8106, 2014.

[Lov16] Shachar Lovett. Communication is bounded by root of rank. *Journal of the ACM*, 63(1):1:1–1:9, 2016. Preliminary version in STOC 2014.

[LS88] László Lovász and Michael Saks. Lattices, Möbius functions and communication complexity. In *Proceedings of the 29th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 81–90, 1988.

[LS09] Troy Lee and Adi Shraibman. Lower bounds in communication complexity. *Foundations and Trends in Theoretical Computer Science*, 3(4):263–399, 2009.

[LW49] Lynn H. Loomis and Hassler Whitney. An inequality related to the isoperimetric inequality. *Bulletin of the American Mathematical Society*, 55:961–962, 1949.

[MS82] Kurt Mehlhorn and Erik M. Schmidt. Las Vegas is better than determinism in VLSI and distributed computing. In *Proceedings of the 14th Annual ACM Symposium on Theory of Computing (STOC)*, pages 330–337, 1982.

[New91] Ilan Newman. Private vs. common random bits in communication complexity. *Information Processing Letters*, 39(2):67–71, 1991.

[NW95] Noam Nisan and Avi Wigderson. On rank vs. communication complexity. *Combinatorica*, 15(4):557–565, 1995.

[Raz92] Alexander A. Razborov. On the distributional complexity of disjointness. *Theoretical Computer Science*, 106(2):385–390, 1992.

[RY20] Anup Rao and Amir Yehudayoff. *Communication Complexity and Applications*. Cambridge University Press, 2020.

[Sha48] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

[ST25] Benny Sudakov and István Tomon. Matrix discrepancy and the log-rank conjecture. *Mathematical Programming*, 212:567–579, 2025. Also arXiv:2311.18524.

[Vav09] Stephen A. Vavasis. On the complexity of nonnegative matrix factorization. *SIAM Journal on Optimization*, 20(3):1364–1377, 2009.

[Vio15] Emanuele Viola. The communication complexity of addition. *Combinatorica*, 35(6):703–747, 2015.

[vN28] John von Neumann. Zur Theorie der Gesellschaftsspiele. *Mathematische Annalen*, 100:295–320, 1928.

[Yan91] Mihalis Yannakakis. Expressing combinatorial optimization problems by linear programs. *Journal of Computer and System Sciences*, 43(3):441–466, 1991.

[Yao77] Andrew Chi-Chih Yao. Probabilistic computations: Toward a unified measure of complexity. In *Proceedings of the 18th IEEE Symposium on Foundations of Computer Science (FOCS)*, pages 222–227, 1977.

[Yao79] Andrew Chi-Chih Yao. Some complexity questions related to distributive computing (preliminary report). In *Proceedings of the 11th Annual ACM Symposium on Theory of Computing (STOC)*, pages 209–213, 1979.